Unconditional Negativity of the Second-Moment Bias and Off-Line Robustness

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Abstract

Building on the curvature identity $O''(\frac{1}{2}) = -2B$ of [1] and the spectral trace formula $\mathrm{Tr}(\widetilde{T}(\sigma)) = D_{\mathrm{SEL}} - O(\sigma)$ of [2], this paper addresses, *unconditionally*, two of the open problems left by [3]. Paper 7 left open a transfer from the integrated bias to the direct second-moment curvature sum; the present paper does *not* prove that implication. Instead it *bypasses* the bridge, proving the direct second-moment negativity independently from the differentiated Guinand–Weil formula, and it controls hypothetical off-critical zeros. We study the second Gaussian moment of the zero sum. Two objects must be distinguished, the *full Guinand–Weil* second-moment sum (over all non-trivial zeros) and its *on-line ordinate proxy*,

$$Z^{(2)}_{p,\mathrm{GW}}(\varepsilon) = \tfrac{1}{2}\sum_{\rho} h^{(2)}_{p,\varepsilon}((\rho - \tfrac{1}{2})/i), \qquad Z^{(2)}_{p,\mathrm{line}}(\varepsilon) = \sum_{\gamma>0} e^{-\varepsilon^2\gamma^2}\gamma^2\cos(\gamma\log p);$$

they coincide when all zeros lie on the critical line and otherwise differ by an off-line correction $E_p^{\mathrm{off},(2)}(\varepsilon) = Z^{(2)}_{p,\mathrm{GW}} - Z^{(2)}_{p,\mathrm{line}}$. With $B^{\infty}_{\mathrm{GW}}(\kappa,\varepsilon) = \sum_{p\le\kappa}(\log p)^2\, Z^{(2)}_{p,\mathrm{GW}}(\varepsilon)$ and $B^{\infty}_{\mathrm{line}}$ defined from the proxy, differentiating the Gaussian explicit-formula identity in the smoothing parameter yields a second-moment identity in which the diagonal prime self-interaction dominates; from it we prove $B^{\infty}_{\mathrm{GW}}(53,\varepsilon) < 0$ for every $0 < \varepsilon \le 0.05$ (closed-form for $\varepsilon \le 0.004$, certified by interval arithmetic up to the reference width $\varepsilon = 0.05$), and a uniform statement $B^{\infty}_{\mathrm{GW}}(\kappa,\varepsilon) < 0$ for every $\kappa \ge 202$ and $0 < \varepsilon \le \varepsilon_0(\kappa)$ with the explicit threshold $\varepsilon_0(\kappa) = 1/(4e\kappa\sqrt{\log\kappa})$. These sign statements are about the full Guinand–Weil object and assume nothing about zero locations. To pass from B^{∞}_{GW} to the operator curvature $O''(\frac{1}{2}) = -2B^{\infty}_{\mathrm{line}}$, which is built from the on-line ordinates, we bound the off-line correction: a β-uniform magnitude estimate combined with the Platt–Trudgian verified height $H_0 = 3\cdot 10^{12}$ [24] shows that $\sum_{p\le\kappa}(\log p)^2|E_p^{\mathrm{off},(2)}|$ stays below the certified negativity margin M for every smoothing width $\varepsilon \ge \varepsilon_{\mathrm{off}}(\kappa, M, H_0)$; within a certified full-object negativity range this transfers the sign, so for $\varepsilon_{\mathrm{off}}(\kappa, M, H_0) \le \varepsilon \le 0.05$ (in particular $\varepsilon = 0.05$) one has $B^{\infty}_{\mathrm{line}}(53,\varepsilon) < 0$ and $O''(\frac{1}{2}) > 0$. No hypothetical input is used: no Riemann Hypothesis, no Hilbert–Pólya postulate, no random-matrix conjecture; the off-line bound rests on an unconditional finite-height verification, not on RH.

What is proved. The second-moment explicit-formula identity (Theorem 3.2); the archimedean bound $\Gamma_p^{(2)} = O(1)$ (Theorem 4.4); prime-side eigenterm extraction and cross-prime control (Theorem 5.1); $B^{\infty}_{\mathrm{GW}}(53,\varepsilon) < 0$ for $0 < \varepsilon \le 0.05$ (Theorem 6.2); the uniform negativity threshold $\varepsilon_0(\kappa)$ for $\kappa \ge 202$ (Theorem 7.3); off-line robustness in magnitude form (Theorem 8.2); and the operator curvature $O''(\frac{1}{2}) = -2B^{\infty}_{\mathrm{line}} > 0$ for $\varepsilon_{\mathrm{off}} \le \varepsilon \le 0.05$, in particular at $\varepsilon = 0.05$ (Corollary 8.3).

What is numerical. At $(\kappa, \varepsilon, N) = (53, 0.05, 100)$: $B(53, 0.05, 100) = -19\,342.5$; $O''(\frac{1}{2}) = +38\,685.1$; $S_3(53) = 87.4247$; $\delta_{\min}(53) = 0.031749$. The archimedean bound $I_4(\frac{1}{4}) \le 3561.1$ is *proved* (Theorem 4.4) and *certified* in Arb ball arithmetic (`certify_I4`).

What remains open. The Cancellation Hypothesis (A**) of [3]; the Weighted Bias Bridge implication $B^{\infty}_{\text{int}} < 0 \Rightarrow B^{\infty}_{\text{GW}} < 0$ (bypassed here, not proved); exact stationarity $O'(\frac{1}{2}) = 0$; the Weil-transfer operator and Weil positivity; and the asymptotics of the negativity window.

Scope. The negativity $B^{\infty}_{\text{GW}} < 0$ of the full Guinand–Weil object, and $B^{\infty}_{\text{line}} < 0$ of the on-line proxy for $\varepsilon_{\text{off}} \le \varepsilon \le 0.05$, are statements about the second-moment bias and the curvature $O''(\frac{1}{2}) = -2B^{\infty}_{\text{line}} > 0$. It is *not* a criterion for the Riemann Hypothesis; this paper does not claim and does not suggest RH (see the scope remark in § 9).

Structure. Sections 3–5 establish the second-moment identity and the three side estimates; Sections 6–7 prove negativity at the reference parameter and uniformly in the cutoff; Section 8 proves off-line robustness. All results are unconditional (Theorem 8.2 is a magnitude statement); no use of RH, GUE, or the Hilbert–Pólya postulate is made.

Notation and Conventions

Critical line. Throughout, the critical line means $\Re(s) = \frac{1}{2}$. The trace parameter $\sigma \in \mathbb{R}$ is evaluated at $\sigma = \frac{1}{2}$ unless stated otherwise.

Global parameters. The reference values are $\kappa = 53$ (prime cutoff, yielding 16 active primes $p \le \kappa$), $\varepsilon = 0.05$ (Gaussian regularisation parameter), and $N = 100$ (number of Riemann ordinates γ_k employed). These values are fixed throughout unless explicitly varied. We write $P(\kappa) := \sum_{p \le \kappa} \log p = \vartheta(\kappa)$ for the Chebyshev theta sum and $S_3(\kappa) := \sum_{p \le \kappa} (\log p)^3/\sqrt{p}$. At the reference parameters $P(53) \approx 44.93$ and $S_3(53) = 87.4247$.

Main objects. The coupling operator $\Phi(\sigma) : H_{\text{str}} \to H_{\text{null}}$ is defined by

$$\Phi(\sigma)_{k,p} = e^{-\varepsilon^2 \gamma_k^2/2} \sin(\sigma \gamma_k \log p),$$

and the Tehrani operator is $\widetilde{T}(\sigma) = \Phi(\sigma) \circ \Phi(\sigma)^*$ ([2]). The trace oscillatory sum is

$$O(\sigma) = \tfrac{1}{2} \sum_{k,p} e^{-\varepsilon^2 \gamma_k^2} \cos(2\sigma \gamma_k \log p),$$

with curvature identity $O''(\frac{1}{2}) = -2B$ ([1]), where

$$B = \sum_{k,p} e^{-\varepsilon^2 \gamma_k^2} (\gamma_k \log p)^2 \cos(\gamma_k \log p)$$

is the finite on-line (ordinate) curvature sum. The present paper works with two infinite-N second-moment objects that must be kept distinct. The *full Guinand–Weil* second-moment zero side and its *on-line ordinate proxy* are

$$Z^{(2)}_{p,\mathrm{GW}}(\varepsilon) := \tfrac{1}{2}\sum_{\rho} h^{(2)}_{p,\varepsilon}\Big(\tfrac{\rho-\frac{1}{2}}{i}\Big), \qquad Z^{(2)}_{p,\mathrm{line}}(\varepsilon) := \sum_{\substack{\rho=\frac{1}{2}+i\gamma\\ \gamma>0}} e^{-\varepsilon^2\gamma^2}\,\gamma^2\cos(\gamma\log p),$$

the sum in $Z^{(2)}_{p,\mathrm{GW}}$ ranging over all non-trivial zeros ρ with multiplicity; their difference is the off-line correction $E^{\mathrm{off},(2)}_p(\varepsilon) := Z^{(2)}_{p,\mathrm{GW}}(\varepsilon) - Z^{(2)}_{p,\mathrm{line}}(\varepsilon)$, which vanishes when all zeros lie on the critical line. The corresponding curvature sums are

$$B^{\infty}_{\mathrm{GW}}(\kappa,\varepsilon) := \sum_{p\le\kappa}(\log p)^2\, Z^{(2)}_{p,\mathrm{GW}}(\varepsilon), \qquad B^{\infty}_{\mathrm{line}}(\kappa,\varepsilon) := \sum_{p\le\kappa}(\log p)^2\, Z^{(2)}_{p,\mathrm{line}}(\varepsilon),$$

so that $B^{\infty}_{\mathrm{line}}$ is the infinite-N form of B, and $O''(\frac{1}{2}) = -2B^{\infty}_{\mathrm{line}}$. At the reference parameters the finite-N curvature sum is $B(53, 0.05, 100) = -19\,342.5$, which agrees with $B^{\infty}_{\mathrm{line}}(53, 0.05)$ up to the truncation control quoted from [1]; we use it as the finite-N numerical proxy for $B^{\infty}_{\mathrm{line}}(53, 0.05)$. The Gaussian test functions are

$$h^{(0)}_{p,s}(u) = e^{-su^2}\cos(u\log p), \qquad h^{(2)}_{p,\varepsilon}(u) = u^2 e^{-\varepsilon^2u^2}\cos(u\log p) = -\,\partial_s h^{(0)}_{p,s}(u)\big|_{s=\varepsilon^2},$$

both even and entire. We write Λ for the von Mangoldt function and $\psi_R(t) := \Re\,\psi(\frac{1}{4}+\frac{it}{2})$ for the real part of the digamma function on the relevant vertical line; the archimedean second-moment term is $\Gamma^{(2)}_p(\varepsilon) = \frac{1}{2\pi}\int_0^\infty t^2 e^{-\varepsilon^2t^2}\cos(t\log p)\,\psi_R(t)\,dt$.

Analytical foundations. The proofs use the Guinand–Weil explicit formula for Gaussian test functions in the form of [19, Thm. 5.12] (the historical sources are [17, 18]), the Riemann–von Mangoldt zero-counting asymptotics $N(T) \sim \frac{T}{2\pi}\log\frac{T}{2\pi}$ in the explicit form of [22], and the Chebyshev-type bound $\psi(x) \le 1.04\,x$ of [21]. Section 8 uses, as a finite verified arithmetic fact, the unconditional interval-arithmetic computation of Platt and Trudgian [24]: every non-trivial zero $\beta+i\gamma$ with $0<\gamma\le H_0 = 3\cdot10^{12}$ has $\beta=\frac{1}{2}$. No zero-location, pair-correlation, or spectral hypothesis is used.

Numerical computations. The illustrative tables and calibration values of this paper (anchors such as $B(53, 0.05, 100) = -19\,342.5$, the residual table, the $\kappa = 202$ budgets) were produced with `mpmath` at 50 decimal digits and serve as candidate values, not as certificates; Riemann zeros γ_k were evaluated via `mpmath.zetazero`. The word *certified* is reserved for the threshold statements of § 6–7, which are established by a separate rigorous *interval* computation in Arb ball arithmetic (midpoint–radius with directed outward rounding) [28], as provided by the FLINT 3 library (Arb layer) through its Python bindings, with proven Gaussian tail bounds below 10^{-380}. The certificate recomputes every prime-power interval contribution, bounds the discarded tail, and prints lower bound, right-hand side, and margin for each covering interval, aborting on any overlap, gap, or non-positive margin. Verification code (the consistency gate and the interval certificate) is available at [4] (`https://github.com/utehrani/analysislab-nt`).

Not to be confused with. The following symbols look similar but are distinct objects.

$Z^{(2)}_{p,\mathrm{GW}}$ **vs.** $Z^{(2)}_{p,\mathrm{line}}$
: full Guinand–Weil zero side $\frac{1}{2}\sum_\rho$ versus the on-line ordinate proxy $\sum_{\gamma>0}$; their difference is $E_p^{\mathrm{off},(2)} = Z^{(2)}_{p,\mathrm{GW}} - Z^{(2)}_{p,\mathrm{line}}$.

B^∞_{GW} **vs.** B^∞_{line}
: full-object sign (this paper) versus the proxy curvature sum (the finite-N ordinate computation gives $-19\,342.5$ at $N = 100$, consistent with the sign obtained after the off-line transfer).

$B^\infty_{\mathrm{GW}}, B^\infty_{\mathrm{line}}$ **vs.** B_{int}
: the γ_k^2-weighted curvature sums versus the integrated bias: the finite reference value is $B^+_{\mathrm{int}}(0.05, 100) = -42.21$ ([1, 3]), distinct from the infinite integrated-bias window B^∞_{int} ([3]).

pole $-\frac{1}{8}$ **vs. ratio** $-\frac{1}{4}$
: the pole coefficient of $2e^{\varepsilon^2/4}\cosh(\frac{\log p}{2})$ is $-\frac{1}{8}$; the ratio to the zeroth-moment pole term is $-\frac{1}{4}$.

$\varepsilon_0(\kappa)$ **vs.** $\varepsilon_{\mathrm{int}}(\kappa)$
: the uniform threshold $1/(4e\kappa\sqrt{\log\kappa})$ of this paper versus the integrated-window scale of [3].

$\varepsilon \le 0.004$ **vs.** $\varepsilon \le 0.05$
: the closed-form threshold versus the certified threshold.

Both pole numbers are correct: the pole term of the second-moment identity equals $-\frac{1}{8} \cdot 2e^{\varepsilon^2/4}\cosh(\frac{\log p}{2})$, i.e. exactly minus one quarter of the zeroth-moment pole term. The closed-form and certified thresholds in Theorem 6.2 are stated separately and never conflated.

Object summary.

Object	Definition	Role
$Z^{(2)}_{p,\mathrm{GW}}$	$\frac{1}{2}\sum_\rho h^{(2)}_{p,\varepsilon}((\rho-\frac{1}{2})/i)$	full GW zero side
$Z^{(2)}_{p,\mathrm{line}}$	$\sum_{\gamma>0} e^{-\varepsilon^2\gamma^2}\gamma^2\cos(\gamma\log p)$	on-line proxy
$E_p^{\mathrm{off},(2)}$	$Z^{(2)}_{p,\mathrm{GW}} - Z^{(2)}_{p,\mathrm{line}}$	off-line correction
B^∞_{GW}	$\sum_{p\le\kappa}(\log p)^2\, Z^{(2)}_{p,\mathrm{GW}}$	proved negative
B^∞_{line}	$\sum_{p\le\kappa}(\log p)^2\, Z^{(2)}_{p,\mathrm{line}}$	operator curvature
$B(\kappa,\varepsilon,N)$	finite ordinate truncation	numerical proxy

1 Introduction

1.1 Background and Motivation

This paper is the eighth in a series developing an operator-theoretic framework for studying the distribution of the non-trivial zeros of the Riemann zeta function, in which prime numbers mediate the coupling between zeros through an explicit family of finite-dimensional operators.

Paper 1 [5] introduced a local curvature decomposition of the explicit formula and identified the critical line as a phase boundary through a structural divergence of the curvature at $\sigma = \frac{1}{2}$.

Paper 2 [6] connected this local curvature to the Weil explicit functional through σ-dependent admissible Gaussian test functions, and identified the Weil positivity framework of [7] as the natural target.

Paper 3 [8] made the geometric structure explicit by introducing two finite-dimensional Hilbert spaces, H_{str} over primes and H_{null} over zeros, together with the coupling operator $\Phi : H_{\mathrm{str}} \to H_{\mathrm{null}}$ built from prime–zero phasors, and the loop operator $T = \Phi^* \circ \Phi$.

Paper 4 [9] introduced the dual Tehrani operator $\widetilde{T} = \Phi \circ \Phi^*$ on H_{null} and showed, through a Hilbert–Pólya diagnostic ($r_2 \to 0.16$ as $\kappa \to \infty$ [numerical]), that $\widetilde{T}$ is *not* a Hilbert–Pólya operator, distinguishing it from the constructions of Berry and Keating [10] and de Branges [11].

Paper 5 [2] extended the operator family to a one-parameter family $\widetilde{T}(\sigma) = \Phi(\sigma) \circ \Phi(\sigma)^*$ and proved the exact spectral trace identity $\mathrm{Tr}(\widetilde{T}(\sigma)) = D_{\mathrm{SEL}} - O(\sigma)$, where D_{SEL} is σ-independent. It introduced the Guinand–Weil smoothed zero sum as motivation for the bias of prime–zero sums.

Paper 6 [1] proved the curvature–bias identity $O''(\frac{1}{2}) = -2B$ by direct differentiation of the trace formula, with $B = \sum_{k,p} e^{-\varepsilon^2\gamma_k^2}(\gamma_k \log p)^2 \cos(\gamma_k \log p)$, and recorded the reference value $B = -19\,342.5$, giving $O''(\frac{1}{2}) = +38\,685.1 > 0$ [numerical]. The transfer from the integrated bias to the direct curvature sum was left open; the present paper bypasses it (see below).

Paper 7 [3] proved, under two named hypotheses, the pointwise asymptotic rate $r_p^\infty = \frac{1}{2}$ and a conditional window $B^\infty_{\mathrm{int}}(\kappa, \varepsilon) < 0$, and formulated the *Weighted Bias Bridge* — the transfer from the integrated bias B^∞_{int} to the direct second-moment sum B^∞_{GW} — and the question of off-critical control as its two central open problems.

The present paper addresses both of these open problems *unconditionally*, by bypassing the bridge rather than establishing it: we do not prove the implication $B^\infty_{\mathrm{int}} < 0 \Rightarrow B^\infty_{\mathrm{GW}} < 0$, but prove $B^\infty_{\mathrm{GW}} < 0$ directly. We work directly with the second-moment object $B^\infty_{\mathrm{GW}}(\kappa, \varepsilon)$, defined through the zero side of the Guinand–Weil explicit formula and hence free of any zero-location hypothesis. The mechanism is elementary in outline: differentiating the Gaussian explicit-formula identity once in the smoothing parameter $s = \varepsilon^2$ produces a second-moment identity whose dominant term, for small ε, is the diagonal self-interaction of each prime with its own term in the explicit formula. This term is negative and of size ε^{-3}; every competing contribution — the pole term, the cross-prime interactions, and the archimedean integral — is of strictly lower order, with explicit constants. The negativity of B^∞_{GW} for small ε is therefore an unconditional, purely prime-side phenomenon.

The operator $\widetilde{T}(\sigma)$ of this series is a Gram-type operator with explicit arithmetic matrix entries; it is not a Hilbert–Pólya operator ([9]), and does not arise from an adelic or non-commutative geometric construction. Connes and Consani [12] and Connes, Consani, and Moscovici [13] construct spectral triples and zeta-cycles on adelic spaces; our framework is finite-cutoff, explicit, and classical. Here B^{∞}_{GW} is an analytic bias object — the second derivative in σ of the spectral trace — not an adelic structure. The methodology of replacing a zero-location hypothesis by a certified finite-height verification combined with explicit-formula estimates is itself well established; the closest mean-square neighbour is the work of Brent, Platt, and Trudgian [25] on the error term in the prime number theorem, and related Hilbert-space approaches to the Weil distribution have been pursued by Burnol [15] and Suzuki [16]. The bridge to the Weil positivity criterion of [7, 14] remains the principal open problem of the series.

1.2 Main Results

The six main results of this paper are as follows.

The first, in § 3 (Theorem 3.2), is the *second-moment identity*: an explicit decomposition of $Z^{(2)}_{p,\mathrm{GW}}(\varepsilon)$ into a pole term with coefficient $-\frac{1}{8}$ (equivalently, minus one quarter of the zeroth-moment pole term), a full prime-power term over all q^m, and an archimedean integral, obtained by differentiating the Gaussian explicit formula in $s = \varepsilon^2$.

The second, in § 4 (Theorem 4.4), is the *archimedean bound* $\Gamma^{(2)}_p(\varepsilon) = O(1)$, with an explicit constant $I_4 \le 3561.1$ obtained by four-fold integration by parts against $\cos(t \log p)$ using $g'''(0) = 0$; this term is subdominant by three powers of ε.

The third, in § 5 (Theorem 5.1), is the *prime-side control*: the diagonal $(q, m) = (p, 1)$ term contributes exactly $\log p/(\sqrt{p}\, 8\sqrt{\pi}\,\varepsilon^3)$, and the cross-prime remainder is controlled through the binding near-zone gap $\delta_{\min}(53) = 0.031749$ (attained by the pair $31 \leftrightarrow 2^5$).

The fourth, in § 6 (Theorem 6.2), is *negativity at the reference parameter*: $B^{\infty}_{\mathrm{GW}}(53, \varepsilon) < 0$ for all $0 < \varepsilon \le 0.05$. The closed-form criterion proves this for $\varepsilon \le 0.004$; a certification by interval arithmetic with the same proven constants extends it to the reference width $\varepsilon = 0.05$ (and, by chaining interval certificates, to $\varepsilon \le 0.0819$). These two stages are stated separately. This certifies the sign of the full Guinand–Weil object; the operator curvature $O''(\frac{1}{2}) = -2B^{\infty}_{\mathrm{line}} > 0$, built from the on-line ordinates, follows once the off-line correction is bounded (Corollary 8.3, after Theorem 8.2).

The fifth, in § 7 (Theorem 7.3), is *uniform negativity*: for every cutoff $\kappa \ge 202$ and every $0 < \varepsilon \le \varepsilon_0(\kappa)$ with $\varepsilon_0(\kappa) = 1/(4e\kappa\sqrt{\log \kappa})$, one has $B^{\infty}_{\mathrm{GW}}(\kappa, \varepsilon) < 0$. This establishes the direct second-moment negativity that the Weighted Bias Bridge would have yielded, but proves it directly rather than through the bridge.

The sixth, in § 8 (Theorem 8.2), is *off-line robustness* in magnitude form: a β-uniform bound on the contribution of any hypothetical off-critical zeros, combined with the Platt–Trudgian verified height $H_0 = 3 \cdot 10^{12}$ and an explicit discrete zero-count inequality, shows that the off-line correction $E_p^{\mathrm{off},(2)}$ stays below the certified negativity margin M for $\varepsilon \geq \varepsilon_{\mathrm{off}}$, without any RH assumption. Within a certified negativity range this transfers the sign from B^{∞}_{GW} to $B^{\infty}_{\mathrm{line}}$, so for $\varepsilon_{\mathrm{off}} \leq \varepsilon \leq 0.05$ (in particular $\varepsilon = 0.05$) one has $O''(\frac{1}{2}) > 0$ (Corollary 8.3).

Relation to existing work. The Guinand–Weil explicit formula, its prime-power and archimedean (digamma) components, the broader literature in which signs are read off from zeros, and the general strategy of removing an RH assumption through a certified finite-height zero verification are all classical; the Gaussian/Hermite-weighted identity technology appears in nearby form in recent work [20]. What is new here is the conjunction: the unconditional pointwise sign $B^{\infty}_{\mathrm{GW}} < 0$ for this specific Gaussian second-moment curvature functional, the constant-tracked archimedean bound (four-fold integration by parts, $I_4 \leq 3561.1$) tailored to the twice-differentiated identity, and the transfer of the certified-zero methodology to off-line robustness of a curvature sign statement. The identity of § 3 is a specialized differentiated descendant of the classical explicit formula, not a new formula; the smoothed and weighted explicit-formula framework, the prime-power/digamma decomposition, and the certified-zero method itself are cited here as prior art, not claimed as contributions.

1.3 Reader Contract

This paper *proves* (unconditionally): the second-moment explicit-formula identity (Theorem 3.2); the archimedean bound $\Gamma_p^{(2)} = O(1)$ (Theorem 4.4); prime-side eigenterm extraction and cross-prime control (Theorem 5.1); $B^{\infty}_{\mathrm{GW}}(53, \varepsilon) < 0$ for all $0 < \varepsilon \leq 0.05$ (Theorem 6.2); the uniform threshold $B^{\infty}_{\mathrm{GW}}(\kappa, \varepsilon) < 0$ for $\kappa \geq 202$, $\varepsilon \leq \varepsilon_0(\kappa)$ (Theorem 7.3); off-line robustness in magnitude form (Theorem 8.2); and the operator curvature positivity $O''(\frac{1}{2}) = -2B^{\infty}_{\mathrm{line}} > 0$ for $\varepsilon_{\mathrm{off}} \leq \varepsilon \leq 0.05$, in particular at $\varepsilon = 0.05$ (Corollary 8.3).

At the reference parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$ this paper *establishes numerically* $B(53, 0.05, 100) = -19\,342.5$, $O''(\frac{1}{2}) = +38\,685.1$, and $\delta_{\min}(53) = 0.031749$. It *proves* the archimedean bound $I_4 \leq 3561.1$ and the closed-form threshold $\varepsilon_0(53) = 0.004$, and *certifies* (interval arithmetic) negativity for $\varepsilon \leq 0.05$ with single-interval margins 0.83 on $(0, 0.047]$ and 24.29 on $[0.047, 0.05]$.

This paper *leaves open*: the Cancellation Hypothesis (A**) of [3]; the Weighted Bias Bridge implication $B^{\infty}_{\mathrm{int}} < 0 \Rightarrow B^{\infty}_{\mathrm{GW}} < 0$ itself (the present paper bypasses it, proving $B^{\infty}_{\mathrm{GW}} < 0$ directly, rather than establishing the implication); exact stationarity $O'(\frac{1}{2}) = 0$; the Weil-transfer operator and Weil positivity; and the asymptotics of the negativity window $\inf_{\kappa} \varepsilon_{\mathrm{int}}(\kappa)$.

This paper does *not* claim and does *not* suggest the Riemann Hypothesis. The negativity $B^{\infty}_{\mathrm{GW}} < 0$ is a second-moment bias / curvature statement, not an RH criterion: the canonical off-critical continuation of the curvature test function yields no sign-definite δ^2-defect, as recorded in the scope remark of § 9.

2 Background: Trace Formula, Curvature Identity, and the Guinand–Weil Framework

This short section records the inputs imported from the preceding papers; all are cited, none re-proved here.

2.1 Trace formula and curvature identity

Paper 5 [2] proves the exact identity $\mathrm{Tr}(\widetilde{T}(\sigma)) = D_{\mathrm{SEL}} - O(\sigma)$ with D_{SEL} σ-independent and $O(\sigma) = \frac{1}{2}\sum_{k,p} e^{-\varepsilon^2\gamma_k^2}\cos(2\sigma\gamma_k \log p)$. Differentiating twice in σ and evaluating at $\sigma = \frac{1}{2}$, Paper 6 [1] obtains the curvature identity

$$O''(\tfrac{1}{2}) = -2B, \qquad B = \sum_{k,p} e^{-\varepsilon^2\gamma_k^2}(\gamma_k \log p)^2 \cos(\gamma_k \log p).$$

The infinite-N, on-line form of B is $B^{\infty}_{\mathrm{line}}(\kappa,\varepsilon) = \sum_{p\le\kappa}(\log p)^2 Z^{(2)}_{p,\mathrm{line}}(\varepsilon)$, so that $O''(\frac{1}{2}) = -2B^{\infty}_{\mathrm{line}}$ and the sign of $B^{\infty}_{\mathrm{line}}$ is the sign of $-O''(\frac{1}{2})$. The present paper proves negativity for the full Guinand–Weil object $B^{\infty}_{\mathrm{GW}}(\kappa,\varepsilon) = \sum_{p\le\kappa}(\log p)^2 Z^{(2)}_{p,\mathrm{GW}}(\varepsilon)$ on the prime/pole/archimedean side (no zero-location hypothesis), and then transfers the sign to $B^{\infty}_{\mathrm{line}}$ by bounding $E_p^{\mathrm{off},(2)}$ in § 8.

Proposition 2.1 (Infinite-N curvature identity; proved)**.** *Write*

$$O_\infty(\sigma) = \tfrac{1}{2}\sum_{k,p} e^{-\varepsilon^2\gamma_k^2}\cos(2\sigma\gamma_k \log p)$$

for the $N \to \infty$ on-line trace. For each fixed $\varepsilon > 0$ the series for O_∞ and its first two σ-derivatives converge locally uniformly in σ, so $O_\infty \in C^2$ and term-by-term differentiation is valid; in particular

$$O''_\infty(\tfrac{1}{2}) = -2B^{\infty}_{\mathrm{line}}(\kappa,\varepsilon).$$

Proof. The ν-th σ-derivative of the (k,p) term is bounded by $e^{-\varepsilon^2\gamma_k^2}(2\gamma_k \log p)^\nu$. By Riemann–von Mangoldt, $N(T) = O(T\log T)$, so for fixed p the ordinate sum $\sum_k e^{-\varepsilon^2\gamma_k^2}\gamma_k^\nu$ is dominated by

$$\int_0^\infty t^\nu e^{-\varepsilon^2 t^2}\, dN(t) = O\Big(\int_0^\infty t^{\nu+1}\log t\, e^{-\varepsilon^2t^2}\, dt\Big) < \infty,$$

the Gaussian damping overpowering the $O(T\log T)$ growth for every fixed $\varepsilon > 0$ and $\nu \le 2$. The bound is independent of σ, so by the Weierstrass M-test each of $O_\infty, O'_\infty, O''_\infty$ converges uniformly on compact σ-sets; differentiation term by term is therefore legitimate, and evaluating the second derivative at $\sigma = \frac{1}{2}$ gives $-2\sum_{k,p} e^{-\varepsilon^2\gamma_k^2}(\gamma_k\log p)^2\cos(\gamma_k \log p) = -2B^{\infty}_{\mathrm{line}}$. □

2.2 The Guinand–Weil framework and its second moment

For an admissible even test function h the Guinand–Weil explicit formula [19, Thm. 5.12] relates the zero side $\frac{1}{2}\sum_\rho h\big((\rho-\frac{1}{2})/i\big)$ (sum over all non-trivial zeros ρ with multiplicity) to a pole term, a prime-power sum against the Fourier transform $\widehat{h}$, and an archimedean integral against the digamma function. When all zeros lie on the critical line the zero side equals $\sum_{k\ge1} h(\gamma_k)$; this is the form used in all numerical computations, the zeros entering the computations ($\gamma_k \le 1002$) being classically known to lie on the line. *No zero-location hypothesis enters any proof below*: all zero sums are defined through the zero side of the explicit formula.

The *second moment* arises from the second σ-derivative of the trace, equivalently from differentiating the even Gaussian test function $h^{(0)}_{p,s}(u) = e^{-su^2}\cos(u\log p)$ once in s at $s = \varepsilon^2$: $h^{(2)}_{p,\varepsilon} = -\partial_s h^{(0)}_{p,s}|_{s=\varepsilon^2}$. This is the object differentiated throughout § 3.

Remark. *The full Guinand–Weil object and the ordinate proxy differ, when off-critical zeros are present, by the off-line contribution $E_p^{\mathrm{off},(2)}$ isolated in § 8; under the all-zeros-on-line identity of Paper 5 they coincide. The present paper never assumes this identity: §§ 3–7 work on the prime/pole/archimedean side, and § 8 bounds $E_p^{\mathrm{off},(2)}$ explicitly.*

3 The Second-Moment Identity

We first record absolute convergence, then the explicit identity. The differentiated Gaussian identity below belongs to the same family of weighted explicit-formula identities as the Gaussian/Hermite-weighted zero sums of [20]; that work is, however, conditional on the Riemann Hypothesis and simplicity of zeros and aims at localization, whereas the unconditional sign statement proved here is of a different nature.

Lemma 3.1 (Absolute convergence of the second moment; proved)**.** *For every $\varepsilon > 0$ and every prime p the sum defining $Z^{(2)}_{p,\mathrm{GW}}(\varepsilon)$ converges absolutely, and so does the full zero-side sum $\frac{1}{2}\sum_\rho h^{(2)}_{p,\varepsilon}((\rho-\frac{1}{2})/i)$.*

Proof. Write $\rho = \beta + i\gamma$ with $0 < \beta < 1$ and $z = (\rho - \frac{1}{2})/i = \gamma - i(\beta - \frac{1}{2})$. Then $\Re(z^2) = \gamma^2 - (\beta-\frac{1}{2})^2 \ge \gamma^2 - \frac{1}{4}$ and $|\cos(zL)| \le e^{|\Im z|L} \le \sqrt{p}$ with $L = \log p$, so

$$\big|h^{(2)}_{p,\varepsilon}(z)\big| \le \big(\gamma^2 + \tfrac{1}{4}\big)\, e^{\varepsilon^2/4}\sqrt{p}\, e^{-\varepsilon^2\gamma^2}.$$

By the Riemann–von Mangoldt formula in the explicit form $N(T) = F(T) + O(\log T)$ with $F(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8}$ and the quantitative remainder $|N(T) - F(T)| \le E(T) = O(\log T)$ of [22] (used quantitatively in § 8), the number of zeros with $n \le |\gamma| < n+1$ is

$$N(n+1) - N(n) \le \big(F(n+1) - F(n)\big) + E(n+1) + E(n) = O(\log(n+2)),$$

since the differenced main term $F(n+1) - F(n) = \frac{1}{2\pi}\log\frac{n}{2\pi} + O(1)$; hence the sum is dominated by $\sum_{n\ge0}(n+1)^2\log(n+2)e^{-\varepsilon^2 n^2} < \infty$. The ordinate sum is the special case $\beta = \frac{1}{2}$. $\square$

Theorem 3.2 (Second-moment explicit-formula identity; proved)**.** *For every prime p and every $\varepsilon > 0$,*

$$Z^{(2)}_{p,\mathrm{GW}}(\varepsilon) = -\tfrac{1}{4}\, e^{\varepsilon^2/4}\cosh\!\Big(\tfrac{\log p}{2}\Big) \;-\; P^{(2)}_p(\varepsilon) \;+\; \Gamma^{(2)}_p(\varepsilon) \;-\; \tfrac{1}{2}(\log\pi)\, A_p(\varepsilon), \qquad (1)$$

where, with $b_\varepsilon(\delta) := \frac{1}{2\varepsilon^2} - \frac{\delta^2}{4\varepsilon^4}$ *and* $L = \log p$,

$$P_p^{(2)}(\varepsilon) = \sum_{q,\,m\ge 1} \frac{\log q}{q^{m/2}} \frac{\sqrt{\pi}}{4\pi\varepsilon}\Big[b_\varepsilon(L - m\log q)\, e^{-(L-m\log q)^2/4\varepsilon^2} + b_\varepsilon(L + m\log q)\, e^{-(L+m\log q)^2/4\varepsilon^2}\Big],$$

$$\Gamma_p^{(2)}(\varepsilon) = \frac{1}{2\pi}\int_0^\infty t^2 e^{-\varepsilon^2 t^2}\cos(tL)\,\psi_R(t)\,dt, \qquad A_p(\varepsilon) = \frac{1}{2\pi}\int_{-\infty}^\infty t^2 e^{-\varepsilon^2 t^2}\cos(tL)\,dt.$$

The pole term equals $-\frac{1}{8}\cdot 2e^{\varepsilon^2/4}\cosh(L/2)$, *i.e. minus one quarter of the zeroth-moment pole term, and is* $O(\sqrt{p})$ *uniformly for* $\varepsilon \le 1$. *The conductor term* $-\frac{1}{2}(\log\pi)A_p(\varepsilon)$ *comes from the factor* $\pi^{-s/2}$ *of the completed zeta function (the half-sum* $\frac{1}{2}\sum_\rho$ *carries the prefactor* $\frac{1}{2}$*); with* $A_p(\varepsilon) = \frac{1}{4\sqrt{\pi}\,\varepsilon^3}\big(1 - \frac{L^2}{2\varepsilon^2}\big)e^{-L^2/4\varepsilon^2}$ *it is* $O\big(\varepsilon^{-5}e^{-L^2/4\varepsilon^2}\big)$ *as* $\varepsilon \downarrow 0$*; the conductor contribution* $-\frac{1}{2}(\log\pi)A_p$ *is numerically* $\approx 8.4\cdot 10^{-17}$ *for* $p = 2$ *at* $\varepsilon = 0.05$, *negligible on the scale of the certified margins, but is retained for exactness. In the interval certificate, which lower-bounds* $\mathrm{Cross} = -8\sqrt{\pi}\,\varepsilon^3 B_{\mathrm{GW}}^\infty$, *the conductor appears as* $+(\log\pi)\,h(L^2/4\varepsilon^2)$ *with a* positive *coefficient: the normalisation* $8\sqrt{\pi}\,\varepsilon^3$ *turns* $-\frac{1}{2}(\log\pi)A_p$ *in* N_{GW} *into* $-(\log\pi)\,h$, *and* $\mathrm{Cross} = -N_{\mathrm{GW}}$ *flips this to* $+(\log\pi)\,h$, *since* $\big(1 - \frac{L^2}{2\varepsilon^2}\big)e^{-L^2/4\varepsilon^2} = +h(L^2/4\varepsilon^2)$.

Proof. Step 1 (zeroth moment). The Gaussian test function $h_{p,s}^{(0)}$ is even, entire, and of rapid decay in horizontal strips, hence admissible [19, Thm. 5.12]; for every $s > 0$,

$$\begin{aligned}\tfrac{1}{2}\sum_\rho h_{p,s}^{(0)}\big((\rho - \tfrac{1}{2})/i\big) = e^{s/4}\cosh\!\Big(\tfrac{L}{2}\Big) &- \sum_{q,m}\frac{\log q}{q^{m/2}}\,\widehat{h}_s^{(0)}(m\log q)\\ &+ \frac{1}{2\pi}\int_0^\infty e^{-st^2}\cos(tL)\,\psi_R(t)\,dt - \tfrac{1}{2}(\log\pi)\,A_p^{(0)}(s),\end{aligned} \tag{2}$$

with prime-side transform and conductor integral

$$\widehat{h}_s^{(0)}(u) = \tfrac{1}{4\sqrt{\pi s}}\big[e^{-(L-u)^2/4s} + e^{-(L+u)^2/4s}\big], \qquad A_p^{(0)}(s) = \tfrac{1}{2\pi}\int_{-\infty}^\infty e^{-st^2}\cos(tL)\,dt.$$

The conductor constant $-\log\pi$ from $\pi^{-s/2}$ does *not* cancel. Carried through the half-sum $\frac{1}{2}\sum_\rho$ it contributes $-\frac{1}{2}(\log\pi)A_p^{(0)}(s)$; applying $-\partial_s$ at $s = \varepsilon^2$ then produces the term $-\frac{1}{2}(\log\pi)A_p(\varepsilon)$ of (1).

Step 2 (differentiate in s). Differentiate (2) in s at $s = \varepsilon^2$ and multiply by -1. On the zero side, termwise differentiation is justified by dominated convergence, the derivative of each term being bounded by the summable majorant of Lemma 3.1; this gives $Z_{p,\mathrm{GW}}^{(2)}(\varepsilon)$. On the pole side, $-\partial_s\, e^{s/4}\cosh(L/2) = -\frac{1}{4}e^{s/4}\cosh(L/2)$. On the prime side, for $\delta = L \mp u$,

$$-\partial_s\big[s^{-1/2}e^{-\delta^2/4s}\big] = s^{-1/2}e^{-\delta^2/4s}\big(\tfrac{1}{2s} - \tfrac{\delta^2}{4s^2}\big),$$

which yields the bracket b_ε with prefactor $\frac{1}{4\sqrt{\pi}\varepsilon} = \frac{\sqrt{\pi}}{4\pi\varepsilon}$; termwise differentiation is justified directly, the s-derivative of the (q,m) term being bounded for $s \ge s_0 > 0$ by $\frac{\log q}{q^{m/2}}\,C(s_0)\,(1 + (m\log q)^2)\,e^{-(L-m\log q)^2/4s_0}$, a Gaussian in $m\log q$ that sums absolutely (independently of the later Theorem 5.1). On the archimedean side, differentiation under the integral sign is justified by the dominating function $t^2e^{-s_0t^2}(\log(2+t)+8)$ for $s \ge s_0 > 0$ (Lemma 4.1). This proves (1). □

Remark (Numerical sanity check)**.** *At $p \in \{2,5,11,37\}$ and $\varepsilon = 0.05$, the two sides of the zeroth-moment identity (2) agree to 10^{-15} and the two sides of the differentiated identity (1) to 10^{-13} relative precision. These checks are not used in the proof.*

4 The Archimedean Term

Lemma 4.1 (Digamma bounds on the quarter line; proved)**.** *With $a_n := n + \frac{1}{4}$ and $\tau := t/2$,*

$$\psi_R(t) = -\gamma_E + \sum_{n\geq 0}\Big(\tfrac{1}{n+1} - \tfrac{a_n}{a_n^2+\tau^2}\Big),$$

and for all $t \geq 0$: $|\psi_R(t)| \leq \log(2+t) + 8$ and $|\psi_R'(t)| \leq \frac{20}{1+t}$. Moreover ψ_R is even, so all odd derivatives vanish at $t = 0$, and $\int_0^\infty S_1(t/2)\,dt = \int_0^\infty t^2 S(t/2)\,dt = \frac{\pi}{2}\psi'(\frac{1}{4}) = \frac{\pi}{2}(\pi^2 + 8G) \approx 27.013$, where S_1, S are the trigamma/quadrigamma series and G is Catalan's constant.

Proof. The series follows from $\psi(z) = -\gamma_E + \sum_{n\geq 0}(\frac{1}{n+1} - \frac{1}{n+z})$ at $z = \frac{1}{4} + i\tau$. *Bound on ψ_R'.* Differentiating, $\psi_R'(t) = -\frac{1}{2}\Im\psi'(\frac{1}{4} + i\tau) = \tau S_1(\tau)$ with $S_1(\tau) = \sum_n a_n/(a_n^2 + \tau^2)^2$ (since $\Im\psi'(\frac{1}{4}+i\tau) = -2\tau\sum_n a_n/(a_n^2+\tau^2)^2$; it is $-\frac{1}{2}\Im\psi'$, *not* $\frac{1}{2}\Re\psi'$, consistent with ψ_R even). We bound $\tau S_1(\tau) \leq 20/(1+t)$ in two regions. For $0 \leq t \leq 2$, per-term maximisation ($\max_\tau \tau a_n/(a_n^2+\tau^2)^2 = 9/(16\sqrt{3}\,a_n^2)$ at $\tau = a_n/\sqrt{3}$) gives $\tau S_1(\tau) \leq M_1 := \frac{9}{16\sqrt{3}}\psi'(\frac{1}{4}) < 5.59 \leq \frac{20}{3} \leq \frac{20}{1+t}$. For $t \geq 2$ (so $\tau \geq 1$), comparing the unimodal summand to its integral plus its peak,

$$S_1(\tau) \leq \int_0^\infty \tfrac{a}{(a^2+\tau^2)^2}\,da + \max_a \tfrac{a}{(a^2+\tau^2)^2} = \tfrac{1}{2\tau^2} + \tfrac{9}{16\sqrt{3}\,\tau^3},$$

so $\tau^2 S_1(\tau) \leq \frac{1}{2} + \frac{9}{16\sqrt{3}\,\tau} \leq \frac{1}{2} + \frac{9}{16\sqrt{3}} < 0.825$ for $\tau \geq 1$; hence $\psi_R'(t) = \tau S_1(\tau) \leq 0.825/\tau = 1.65/t < \frac{20}{1+t}$. This proves the envelope on all of $[0,\infty)$.

Bound on ψ_R. Write $z = \frac{1}{4} + i\frac{t}{2}$ and use the recurrence $\psi(z) = \psi(z+1) - 1/z$. Since $\Re(z+1) = \frac{5}{4} \geq \frac{1}{2}$, the standard digamma estimate $|\psi(w) - \log w| \leq 1/|w|$ for $\Re w \geq \frac{1}{2}$ [31, 6.3.18] gives $|\psi(z+1) - \log(z+1)| \leq 1/|z+1| \leq \frac{4}{5}$. Moreover $\Re\log(z+1) = \frac{1}{2}\log(\frac{25}{16} + \frac{t^2}{4}) \leq \log(2+t)$ (as $\frac{25}{16} + \frac{t^2}{4} \leq (2+t)^2$), and $|\Re(1/z)| = \frac{1}{4}/(\frac{1}{16} + \frac{t^2}{4}) \leq 4$. Hence

$$\begin{aligned}|\psi_R(t)| = |\Re\psi(z)| &\leq |\Re\log(z+1)| + |\psi(z+1) - \log(z+1)| + |\Re(1/z)| \\ &\leq \log(2+t) + \tfrac{4}{5} + 4 < \log(2+t) + 8.\end{aligned}$$

Both envelopes are upper bounds on the true functions, so the I_4 assembly (which uses only $|\psi_R'| \leq 20/(1+t)$ and $|\psi_R| \leq \log(2+t) + 8$; the sharp higher-derivative bounds come from Lemma 4.2) remains valid. The exact integrals reduce, after the substitution $t = 2a\lambda$, to $\frac{\pi}{2}\sum_n a_n^{-2} = \frac{\pi}{2}\psi'(\frac{1}{4})$, and $\psi'(\frac{1}{4}) = \pi^2 + 8G$. ∎

For the fourth-derivative bound below we record the exact derivatives of $\varphi(t) := t^2 e^{-\varepsilon^2 t^2}$,

$$\varphi' = (2t - 2\varepsilon^2 t^3)\,e^{-\varepsilon^2 t^2}, \qquad \varphi'' = (2 - 10\varepsilon^2 t^2 + 4\varepsilon^4 t^4)\,e^{-\varepsilon^2 t^2},$$
$$\varphi''' = (-24\varepsilon^2 t + 36\varepsilon^4 t^3 - 8\varepsilon^6 t^5)\,e^{-\varepsilon^2 t^2}, \quad \varphi'''' = (-24\varepsilon^2 + 156\varepsilon^4 t^2 - 112\varepsilon^6 t^4 + 16\varepsilon^8 t^6)\,e^{-\varepsilon^2 t^2},$$

together with the following higher digamma bounds.

Lemma 4.2 (Higher digamma bounds; proved)**.** *Let $S(\tau) := \sum_{n\geq 0} a_n/(a_n^2+\tau^2)^3$ and $S_1(\tau) := \sum_{n\geq 0} a_n/(a_n^2+\tau^2)^2$, with $a_n = n+\frac{1}{4}$ and $\tau = t/2$. Then for all $t \geq 0$,*

$$|\psi_R''(t)| \leq \tfrac{3}{2}S_1(\tau), \qquad |\psi_R'''(t)| \leq 3\tau\, S(\tau), \qquad |\psi_R''''(t)| \leq \tfrac{15}{2}\, S(\tau),$$

and moreover

(i) $\displaystyle\int_0^\infty S_1(t/2)\,dt = \int_0^\infty t^2 S(t/2)\,dt = \tfrac{\pi}{2}\psi'(\tfrac{1}{4}) = \tfrac{\pi}{2}(\pi^2+8G) \approx 27.013$*;*

(ii) *for $\tau \geq 1$, $S(\tau) \leq c_S/\tau^4$ with $c_S = \frac{1}{4} + \frac{125}{216\sqrt{5}} \leq 0.5089$, and $S_1(\tau) \leq c_{S_1}/\tau^2$ with $c_{S_1} = \frac{1}{2} + \frac{3\sqrt{3}}{16} \leq 0.8248$;*

(iii) $\displaystyle\sup_{t>0} t^3 S(t/2) \leq \sum_{n\geq 0} a_n^{-2} = \psi'(\tfrac{1}{4})$*;*

(iv) *$S(\tau) \leq \zeta(5,\frac{1}{4})$ and $S_1(\tau) \leq \zeta(3,\frac{1}{4})$ for all $\tau \geq 0$, both decreasing in τ.*

Proof. Termwise differentiation of the series of Lemma 4.1 is justified by locally uniform convergence; with $f_a(\tau) = a/(a^2+\tau^2)$ and $\frac{d}{dt} = \frac{1}{2}\frac{d}{d\tau}$,

$$f_a'' = \frac{-2a(a^2-3\tau^2)}{(a^2+\tau^2)^3}, \quad f_a''' = \frac{24a\tau(a^2-\tau^2)}{(a^2+\tau^2)^4}, \quad f_a'''' = \frac{24a(a^4-10a^2\tau^2+5\tau^4)}{(a^2+\tau^2)^5}.$$

Using $|a^2-3\tau^2| \leq 3(a^2+\tau^2)$, $|a^2-\tau^2| \leq a^2+\tau^2$, and (with $a^2 = (1-s)u$, $\tau^2 = su$) $\max_{0\leq s\leq 1}|1-12s+16s^2| = 5$, so $|a^4-10a^2\tau^2+5\tau^4| \leq 5(a^2+\tau^2)^2$, dividing by $4, 8, 16$ gives the three pointwise bounds. For (i), substitute $t = 2a\lambda$ and use $\int_0^\infty (1+\lambda^2)^{-2}d\lambda = \pi/4$, $\int_0^\infty \lambda^2(1+\lambda^2)^{-3}d\lambda = \pi/16$; both reduce to $\frac{\pi}{2}\sum_n a_n^{-2}$. For (ii), integral comparison with the maximal summand gives the stated constants; for (iii), $\max_{\tau>0} 8a\tau^3/(a^2+\tau^2)^3 = a^{-2}$ at $\tau = a$; for (iv), each summand decreases in τ and equals the Hurwitz value at $\tau = 0$. □

Lemma 4.3 (Forward bound for the I_4 residual; proved, certified)**.** *Let $T_1(\varepsilon) = \int_0^\infty |\varphi''''|\,|\psi_R|\,dt$ and let $R_3(\varepsilon) = T_3(\varepsilon) - 9\pi\psi'(\frac{1}{4})$, $R_4(\varepsilon) = T_4(\varepsilon) - 6\pi\psi'(\frac{1}{4})$ be the remainders of the two main terms (notation as in Theorem 4.4 below). There are elementary majorants*

$$T_1(\varepsilon) \leq A_1(\varepsilon), \qquad R_3(\varepsilon) \leq A_3(\varepsilon), \qquad R_4(\varepsilon) \leq A_4(\varepsilon),$$

with endpoint values, and with A_1 increasing on $(0,\frac{1}{4}]$ while A_3, A_4 obey ε-uniform bounds (proof),

$$A_1(\tfrac{1}{4}) \leq 475, \quad A_3(\tfrac{1}{4}) \leq 57, \quad A_4(\tfrac{1}{4}) \leq 14,$$

so that

$$A_1(\tfrac{1}{4}) + A_3(\tfrac{1}{4}) + A_4(\tfrac{1}{4}) \leq 546 \leq 1697.1.$$

Proof. Strategy. We bound $|g''''|$ termwise into three majorants A_1, A_3, A_4, certify their values at $\varepsilon = \frac{1}{4}$ in Arb, and then establish the ε-uniform bound on $(0, 0.05]$ separately (no monotonicity).

With $\varphi(t) = t^2 e^{-\varepsilon^2 t^2}$, $\varphi'''' = (-24\varepsilon^2 + 156\varepsilon^4 t^2 - 112\varepsilon^6 t^4 + 16\varepsilon^8 t^6)e^{-\varepsilon^2 t^2}$, so by the triangle inequality and $|\psi_R(t)| \le \log(2+t) + 8$ (Lemma 4.1),

$$A_1(\varepsilon) := \int_0^\infty \bigl(24\varepsilon^2 + 156\varepsilon^4 t^2 + 112\varepsilon^6 t^4 + 16\varepsilon^8 t^6\bigr) e^{-\varepsilon^2 t^2} \bigl(\log(2+t) + 8\bigr)\, dt$$

is a majorant for T_1; each monomial is a Gaussian moment times a logarithmic factor, and the interval certificate (`cert_paper8.py`, `certify_I4`) integrates this majorant rigorously in Arb ball arithmetic, giving $A_1(\frac{1}{4}) = 474.6 \le 475$. For R_3, write $\varphi'' = (2 - 10\varepsilon^2 t^2 + 4\varepsilon^4 t^4)e^{-\varepsilon^2 t^2}$ and $|\psi_R''| \le \frac{3}{2} S_1(t/2)$ (Lemma 4.2); the constant part 2 contributes $9\int 2\, S_1(t/2)\, dt = 9\pi\psi'(\frac{1}{4})$, exactly the subtracted T_3 main term, so

$$R_3(\varepsilon) \le A_3(\varepsilon) := 9\int_0^\infty (10\varepsilon^2 t^2 + 4\varepsilon^4 t^4) e^{-\varepsilon^2 t^2} S_1(t/2)\, dt,$$

with S_1 the sharp series of Lemma 4.2 (not the crude $(1+\tau)^{-2}$ bound); A_3 is integrated rigorously in Arb, $A_3(\frac{1}{4}) = 56.1 \le 57$. For R_4, from $\varphi' = (2t - 2\varepsilon^2 t^3)e^{-\varepsilon^2 t^2}$ and $|\psi_R'''| \le 3\tau S(\tau)$ the leading $2t$ part contributes $12\int t^2 S(t/2)\, dt = 6\pi\psi'(\frac{1}{4})$ (the T_4 main term), leaving

$$R_4(\varepsilon) \le A_4(\varepsilon) := 12\varepsilon^2 \int_0^\infty t^4 e^{-\varepsilon^2 t^2} S(t/2)\, dt,$$

again integrated in Arb, $A_4(\frac{1}{4}) = 13.7 \le 14$. Summing the three endpoint values gives $474.6 + 56.1 + 13.7 = 544.4 \le 1697.1$; the residual bounds A_1, A_3, A_4, the three main terms, and T_2 are all reproduced as Arb outputs by `certify_I4` (which also self-tests that the digamma majorants dominate the true series), so $I_4(\frac{1}{4}) \le 3561.1$ is certified, not merely numerical.

For the certificate range $0 < \varepsilon \le 0.05$ we do *not* invoke any monotonicity of I_4; instead the uniform bound is certified directly (`certify_Gw_uniform`). After the substitution $u = \varepsilon t$,

$$A_1(\varepsilon) = \varepsilon \int_0^\infty (24 + 156u^2 + 112u^4 + 16u^6) e^{-u^2} \bigl(\log(2 + u/\varepsilon) + 8\bigr)\, du,$$

whose integrand carries the factor $\varepsilon\bigl(\log(2+u/\varepsilon)+8\bigr)$ with ∂_ε-derivative $\log(2+u/\varepsilon)+8-\frac{1}{1+2\varepsilon/u} \ge 7 > 0$, so A_1 is increasing and $A_1(\varepsilon) \le A_1(0.05)$. Writing $\tau = u/2\varepsilon$,

$$A_3(\varepsilon) = 18\int_0^\infty (10u + 4u^3) e^{-u^2}\, \tau S_1(\tau)\, du \le 126\, M_1,$$
$$A_4(\varepsilon) = 96\int_0^\infty u\, e^{-u^2}\, \tau^3 S(\tau)\, du \le 48\, M_3,$$

where $\int (10u + 4u^3)e^{-u^2} = 7$, $\int u\, e^{-u^2} = \frac{1}{2}$, and the suprema are bounded by summing the per-term maxima, $M_1 := \sup_{\tau \ge 0} \tau S_1(\tau) \le \frac{9}{16\sqrt{3}}\, \psi'(\frac{1}{4})$ (each $\tau\, a_n/(a_n^2 + \tau^2)^2$ peaks at $\tau = a_n/\sqrt{3}$) and $M_3 := \sup_{\tau \ge 0} \tau^3 S(\tau) \le \frac{1}{8}\, \psi'(\frac{1}{4})$ (peak at $\tau = a_n$). Hence $I_4(\varepsilon) \le T_2(0.05) + (\text{main}) + A_1(0.05) + 126M_1 + 48M_3$, giving $G_w(\varepsilon) \le 1586 \le 2693$ for all $\varepsilon \in (0, 0.05]$ as a single ε-uniform Arb output. It is this uniform bound, not monotonicity, that transfers the archimedean constant to the whole certificate range. □

Theorem 4.4 (Explicit archimedean bound; proved). *For all $0 < \varepsilon \le \frac{1}{4}$ and all primes $p \ge 2$*

the per-prime bound $|\Gamma_p^{(2)}(\varepsilon)| \le I_4(\varepsilon)/2\pi(\log p)^4$ *holds; on the certificate range* $0 < \varepsilon \le 0.05$ *and at the endpoint* $\varepsilon = \frac{1}{4}$ *the certified ceiling* $I_4 \le 3561.1$ *(Lemma 4.3, no monotonicity) gives*

$$\left|\Gamma_p^{(2)}(\varepsilon)\right| \;\le\; \frac{I_4(\varepsilon)}{2\pi(\log p)^4} \;\le\; \frac{567}{(\log p)^4} \;\le\; \frac{1180}{(\log p)^2},$$
$$I_4(\tfrac{1}{4}) \le 3561.1, \quad I_4(\varepsilon) \le 3561.1 \textit{ on } (0, 0.05] \text{ (\textit{Lemma 4.3})}.$$

In particular $\Gamma_p^{(2)}(\varepsilon) = O(1)$*, uniformly in* p*, and the weighted total obeys*

$$\sum_{p \le 53} (\log p)^2 |\Gamma_p^{(2)}(\varepsilon)| \le 2693.$$

Proof. Write $\varphi(t) = t^2 e^{-\varepsilon^2 t^2}$ and $g = \varphi\,\psi_R$, so that $2\pi\Gamma_p^{(2)} = \int_0^\infty g(t)\cos(tL)\,dt$. Since $\varphi(0) = \varphi'(0) = \varphi'''(0) = 0$, $\varphi''(0) = 2$, and all odd derivatives of ψ_R vanish at 0 (Lemma 4.1), one has $g(0) = g'(0) = 0$ and $g'''(0) = 6\,\psi_R'(0) = 0$. Four integrations by parts against $\cos(tL)$ therefore leave no boundary terms and give

$$\int_0^\infty g(t)\cos(tL)\,dt = \frac{1}{L^4}\int_0^\infty g''''(t)\cos(tL)\,dt, \qquad |\Gamma_p^{(2)}| \le \frac{I_4(\varepsilon)}{2\pi L^4}, \quad I_4(\varepsilon) := \int_0^\infty |g''''(t)|\,dt.$$

Assembly of I_4*.* By Leibniz,

$$g'''' = \varphi''''\psi_R + 4\varphi'''\psi_R' + 6\varphi''\psi_R'' + 4\varphi'\psi_R''' + \varphi\,\psi_R'''',$$

and we bound the five integrals $T_1, \dots, T_5$ separately, writing $E := e^{-\varepsilon^2 t^2}$ and $M_j := \int_0^\infty t^j E\,dt = \Gamma(\frac{j+1}{2})/(2\varepsilon^{j+1})$.

- $T_1 = \int |\varphi''''|\,|\psi_R|\,dt$: with $|\psi_R| \le \log(2+t) + 8$ (Lemma 4.1) and $\log(2+t) \le \log(2+1/\varepsilon) + \log(1+\varepsilon t)$, each of the four monomials of φ'''' contributes a term carrying a prefactor ε^{2k+2}, all $O(\varepsilon \log(1/\varepsilon))$.
- $T_2 = 4\int |\varphi'''|\frac{20}{1+t}\,dt \le 80(24\varepsilon^2 M_0 + 36\varepsilon^4 M_2 + 8\varepsilon^6 M_4) = O(\varepsilon)$, using $t^j/(1+t) \le t^{j-1}$.
- $T_3 = 6\int |\varphi''|\frac{3}{2}S_1(t/2)\,dt \le 9\big[2\int_0^\infty S_1 + 10\varepsilon^2 \int t^2 E S_1 + 4\varepsilon^4 \int t^4 E S_1\big]$; the first integral is exact by Lemma 4.2(i), the remainders use the split at $t = 2$ with Lemma 4.2(ii),(iv) and are $O(\varepsilon)$. Main term $18 \cdot \frac{\pi}{2}\psi'(\frac{1}{4}) = 9\pi\psi'(\frac{1}{4}) \approx 486.2$.
- $T_4 = 4\int |\varphi'|\frac{3t}{2}S(t/2)\,dt \le 12\int t^2 E\, S(t/2)\,dt + O(\varepsilon^0)$; on $[0, 2]$ use $t^5 S(t/2) \le t^2\psi'(\frac{1}{4})$ (Lemma 4.2(iii)) and on $[2, \infty)$ the decay $t^5 S(t/2) \le 16c_S t$. Main term $12 \cdot \frac{\pi}{2}\psi'(\frac{1}{4}) = 6\pi\psi'(\frac{1}{4}) \approx 324.2$, remainder $\le \frac{8}{3}\psi'(\frac{1}{4}) \cdot 12\varepsilon^2 + 96c_S \le 34.4 + 48.9$ on $(0, \frac{1}{4}]$.
- $T_5 = \int \varphi\,\frac{15}{2}S(t/2)\,dt \le \frac{15}{2}\int t^2 S(t/2)\,dt = \frac{15\pi}{4}\psi'(\frac{1}{4}) \approx 202.6$ (using $E \le 1$).

Evaluated at $\varepsilon = \frac{1}{4}$ and rounded up, the five rows sum to the conservative endpoint ceiling $I_4(\frac{1}{4}) \le 3561.1$, which the proof uses. The accompanying Arb certificate `certify_I4` recomputes the same archimedean budget and outputs the sharper endpoint enclosure $I_4(\frac{1}{4}) \le 2408.17$ (together with the ε-uniform bound on $(0, 0.05]$ of Lemma 4.3, no monotonicity, used in the weighted total); this sharper enclosure is not needed for the proof, it verifies the conservative ceiling used below. The residual row is bounded *forward* by Lemma 4.3 ($A_1 + A_3 + A_4 \le 544.4$ at $\varepsilon = \frac{1}{4}$), well within the 1697.1 budget, so the table summarizes a proof already present in the bounds rather than introducing 1697.1 by subtraction:

term	majorant at $\varepsilon = \frac{1}{4}$	upper bound
T_3 main $= 9\pi\psi'(\frac{1}{4})$	$9\pi(\pi^2 + 8G)$	≤ 486.3
T_4 main $= 6\pi\psi'(\frac{1}{4})$	$6\pi(\pi^2 + 8G)$	≤ 324.2
$T_5 = \frac{15\pi}{4}\psi'(\frac{1}{4})$	$\frac{15\pi}{4}(\pi^2 + 8G)$	≤ 202.7
T_2	$80(24\varepsilon^2 M_0 + 36\varepsilon^4 M_2 + 8\varepsilon^6 M_4)$	≤ 850.8
T_1 and the T_3, T_4 remainders	forward via Lem. 4.3 (≤ 544.4)	≤ 1697.1 (budget)
$I_4(\frac{1}{4})$	total	≤ 3561.1

Hence $567/L^4 \le 567/(L^2(\log 2)^2) \le 1180/L^2$ as a per-prime bound. For the weighted total the rounded constant $567 = \lceil I_4(\frac{1}{4})/2\pi \rceil$ must be avoided (it would give $567 \cdot 4.7514 = 2694.0 > 2693$); using the unrounded value $I_4(\frac{1}{4})/2\pi = 3561.1/2\pi = 566.77$,

$$\sum_{p \le 53} (\log p)^2 |\Gamma_p^{(2)}(\varepsilon)| \;\le\; \frac{I_4(\frac{1}{4})}{2\pi} \sum_{p \le 53} (\log p)^{-2} \;=\; \frac{3561.1}{2\pi} \cdot 4.7514 \;<\; 2693.$$

$\square$

Remark. *The bound is conservative by roughly two orders of magnitude. Numerically, the weighted terms satisfy* $(\log p)^2 \Gamma_p^{(2)}(\varepsilon) \in [0.26, 0.76]$ *for* $p \le 53$ *and* $\varepsilon \in [0.01, 0.1]$*, and the residuals numerically appear to remain bounded, consistent with finite* $O(1)$ *limiting behaviour as* $\varepsilon \to 0$*. The point is that* $\Gamma_p^{(2)}$ *is subdominant by three powers of* ε *against the prime-side main term of order* ε^{-3}*.*

5 The Prime Side

Theorem 5.1 (Eigenterm extraction and cross-prime control; proved)**.** *Fix* $\kappa \ge 2$ *and a prime* $p \le \kappa$*, and let*

$$\delta_p := \min\{\, |L - m \log q| : q \text{ prime},\ m \ge 1,\ (q, m) \ne (p, 1) \,\} > 0, \qquad L = \log p.$$

Then for all $0 < \varepsilon \le \frac{1}{4}$,

$$P_p^{(2)}(\varepsilon) = \frac{\log p}{\sqrt{p}\, 8\sqrt{\pi}\, \varepsilon^3} \;+\; E_p(\varepsilon), \qquad |E_p(\varepsilon)| \le \frac{c_p}{\varepsilon^5}\, e^{-\delta_p^2/4\varepsilon^2} + \frac{c_p^{\text{far}}}{\varepsilon^3}\, e^{-1/16\varepsilon^2},$$

with explicit constants $c_p = \frac{3\sqrt{\pi}}{16\pi} \sum_{(q,m) \in \mathcal{N}_p} \frac{\log q}{q^{m/2}}$ *(the direct near sum) and* $c_p^{\text{far}} = \frac{5\sqrt{\pi}}{16\pi}(1.506\, p + 1.04\sqrt{p} + 0.76)$*, where* $\mathcal{N}_p = \{(q, m) \ne (p, 1) : |L - m \log q| < 1\}$ *and* $n_p = \#\mathcal{N}_p$*.*

Proof. Eigenterm. The term $(q,m)=(p,1)$ has $\delta_-=0$, hence $b_\varepsilon(0)=\frac{1}{2\varepsilon^2}$ and contributes exactly

$$\frac{\log p}{\sqrt{p}}\cdot\frac{\sqrt{\pi}}{4\pi\varepsilon}\cdot\frac{1}{2\varepsilon^2}=\frac{\log p}{\sqrt{p}\,8\sqrt{\pi}\,\varepsilon^3},$$

with no multiplicative error; its $\delta_+=2L$ companion is absorbed below.

Near zone. For $(q,m)\in\mathcal{N}_p$ one has $\delta_p\le|\delta_-|<1$, so for $\varepsilon\le1$, $|b_\varepsilon(\delta_-)|\le\frac{3}{4\varepsilon^4}$ and $e^{-\delta_-^2/4\varepsilon^2}\le e^{-\delta_p^2/4\varepsilon^2}$; summing the direct near sum $\sum_{(q,m)\in\mathcal{N}_p}\log q/q^{m/2}$ (not n_p max) gives the first bound.

Far zone. For any term with $\delta:=|L\mp m\log q|\ge1$, $|b_\varepsilon(\delta)|e^{-\delta^2/4\varepsilon^2}\le\frac{5}{4\varepsilon^2}e^{-\delta^2/16\varepsilon^2}e^{-1/16\varepsilon^2}$, and for $\varepsilon\le\frac14$, $e^{-\delta^2/16\varepsilon^2}\le e^{-\delta}$. Writing $n=q^m$, the three far families are summed against $e^{-\delta}$: for $n>ep$, $\sum_n\Lambda(n)n^{-3/2}p\le1.506\,p$ (the full Dirichlet series $\sum_n\Lambda(n)n^{-3/2}=-\zeta'(\frac32)/\zeta(\frac32)=1.5052\ldots\le1.506$ is a majorant); for $n<p/e$, $\sqrt{p}\,\psi(p)/p\le1.04\sqrt{p}$ by Chebyshev [21]; for the δ_+ terms, $\frac1p\sum_n\Lambda(n)n^{-3/2}\le0.76$. Multiplying by $\frac{\sqrt{\pi}}{4\pi\varepsilon}$ yields c_p^{far}. $\square$

Remark. *For $\kappa=53$ the binding proximity is $\delta_{\min}(53)=\min_{p\le53}\delta_p=0.031749$, attained by the pair $31\leftrightarrow2^5=32$; the weighted near-zone and far-zone constants are*

$$\sum_{p\le53}(\log p)^2c_p=163.6\ (=\lceil163.611\rceil,\ \textit{direct near sum}),\qquad\sum_{p\le53}(\log p)^2c_p^{\mathrm{far}}\le1385.$$

6 Negativity at the Reference Parameter

Summing the identity of Theorem 3.2 against the weights $(\log p)^2$ over $p\le\kappa$ gives

$$B_{\mathrm{GW}}^\infty(\kappa,\varepsilon)=-M(\varepsilon)+\widetilde{R}(\varepsilon),\qquad M(\varepsilon)=\frac{S_3(\kappa)}{8\sqrt{\pi}\,\varepsilon^3},\quad|\widetilde{R}(\varepsilon)|\le R(\varepsilon),\tag{3}$$

where $-M(\varepsilon)$ collects the diagonal eigenterms of Theorem 5.1 (each contributing $-(\log p)^2\cdot\frac{\log p}{\sqrt{p}8\sqrt{\pi}\varepsilon^3}$, with the overall sign from $B_{\mathrm{GW}}^\infty=\sum(\log p)^2(\cdots-P_p^{(2)}+\Gamma_p^{(2)})$), and $R(\varepsilon)$ bounds, respectively, the weighted near- and far-zone cross-prime errors of Theorem 5.1, the weighted archimedean total of Theorem 4.4, and the weighted pole term:

$$R(\varepsilon)=\frac{C_{\mathrm{near}}(\kappa)}{\varepsilon^5}\,e^{-\delta_{\min}^2/4\varepsilon^2}+\frac{C_{\mathrm{far}}(\kappa)}{\varepsilon^3}\,e^{-1/16\varepsilon^2}+G_w(\kappa)+\frac{e^{1/4}}{4}\sum_{p\le\kappa}(\log p)^2\cosh\tfrac{\log p}{2},$$

with $C_{\mathrm{near}}(\kappa)=\sum_{p\le\kappa}(\log p)^2c_p=\frac{3\sqrt{\pi}}{16\pi}W_{\mathrm{near}}(\kappa)$ (the direct near sum of Lemma 7.2(b), with no n_p max inflation), $C_{\mathrm{far}}(\kappa)=\sum_{p\le\kappa}(\log p)^2c_p^{\mathrm{far}}$, and

$$G_w(\kappa)=\sup_{0<\varepsilon\le0.05}\sum_{p\le\kappa}(\log p)^2|\Gamma_p^{(2)}(\varepsilon)|$$

the (uniform, ε-independent) weighted archimedean total bounded by Theorem 4.4 on the certified range $(0,0.05]$ (≤2693 for $\kappa=53$; the supremum is taken over the range where $I_4\le3561.1$ is certified, not over $(0,\frac14]$). For $\kappa=53$, $C_{\mathrm{near}}(53)=163.6$ (direct sum) and $C_{\mathrm{far}}(53)\le1385$. Hence $B_{\mathrm{GW}}^\infty<0$ whenever $R(\varepsilon)<M(\varepsilon)$. We state the closed-form and certified ranges separately.

6.1 Closed-form core

Theorem 6.1 (Closed-form negativity threshold; proved)**.** *For $\kappa = 53$, $B^{\infty}_{\mathrm{GW}}(53, \varepsilon) < 0$ for all $0 < \varepsilon \le \varepsilon_0(53) := 0.004$. More generally, for every $\kappa \ge 2$ and every $0 < \varepsilon \le \min\{0.05, \delta_{\min}/2\}$ with $R(\varepsilon) < M(\varepsilon)$, and for all smaller ε, one has $B^{\infty}_{\mathrm{GW}}(\kappa, \varepsilon) < 0$.*

Proof. By (3) it suffices that $R(\varepsilon) < M(\varepsilon)$, and that the inequality propagates downward in ε. Each error-to-main ratio is non-decreasing on the stated range: the near-zone ratio is proportional to $\varepsilon^{-2} e^{-\delta^2_{\min}/4\varepsilon^2}$ (increasing for $\varepsilon < \delta_{\min}/2$), the far-zone ratio to $e^{-1/16\varepsilon^2}$, the archimedean ratio to ε^3 (since $G_w(\varepsilon) \le 2693$ holds uniformly, Lemma 4.3, while $M \propto \varepsilon^{-3}$), and the pole ratio to $\varepsilon^3 e^{\varepsilon^2/4}$; all increasing. Hence $R(\varepsilon_0) < M(\varepsilon_0)$ at some admissible ε_0 implies the same for all $0 < \varepsilon \le \varepsilon_0$. For $\kappa = 53$: $S_3(53) = 87.4247$, $\pi(53) = 16$, $\delta_{\min} = 0.031749$, and the weighted constants of § 5, § 4. At $\varepsilon_0 = 0.004 \le \delta_{\min}/2$ one computes $M = 9.63 \cdot 10^7$, near-zone $\le 3.6 \cdot 10^7$, far-zone $\le 10^{-1000}$, archimedean $G_w(53) \le 2693$, pole ≤ 102, so $R < M$ with margin, and the monotonicity step extends the conclusion to all $0 < \varepsilon \le 0.004$. ☐

6.2 Certified extension to the reference width

The closed-form criterion is deliberately conservative: it bounds every near-zone term by the global minimum $\delta_{\min}$ and uses the crude archimedean constant. Replacing the closed-form cross-prime bound by the exact (finitely truncated, with rigorously negligible tail) absolute cross sums, while keeping the proven archimedean constant of Theorem 4.4 and the pole bound, yields the following certified statement.

Theorem 6.2 (Negativity at the reference parameter; proved, certified)**.** $B^{\infty}_{\mathrm{GW}}(53, \varepsilon) < 0$ *for all* $0 < \varepsilon \le 0.05$. *The Arb interval criterion holds on* $[10^{-9}, 0.047]$ *with margin* 0.83 *and on* $[0.047, 0.05]$ *with margin* 24.29*; the remaining* $(0, 10^{-9})$ *is covered by Theorem 6.1, so together these prove* $(0, 0.05]$*. It uses exact finite sums over all prime powers with* $|\log n \mp \log p| < 3$ *in 256-bit (≈77-digit) Arb interval arithmetic. The discarded prime powers are bounded by a Gauss–Chebyshev majorant: grouping them by unit* $\log n$*-bins, the bin count is at most* $\psi(e^{k+1}) \le 1.04\, e^{k+1}$ *(Chebyshev),* $\Lambda(n) \le \log n < k+1$*, and* $|h| \le (2x-1)e^{-x}$ *with* $x = (\log n \mp \log p)^2/4\varepsilon^2$*; the quadratic Gaussian decay* e^{-x} *dominates the single-exponential bin growth, so the tail series is super-exponentially convergent and rigorously below* 10^{-380}.

Proof. Multiply Theorem 3.2 by $8\sqrt{\pi}\varepsilon^3(\log p)^2$ and sum, so the prime side becomes $S_3(53) + \mathrm{Cross}(\varepsilon)$ with the cross terms expressed through $h(x) := (1-2x)e^{-x}$ evaluated at $x_{n,\pm} = (\log n \mp \log p)^2/4\varepsilon^2$. On any interval $[a, b]$ the argument of each x ranges in $[x(b), x(a)]$. In the normalised lower-bound expression for $\mathrm{Cross} = -8\sqrt{\pi}\,\varepsilon^3 B^{\infty}_{\mathrm{GW}}$, both the prime cross terms and the conductor term $+(\log \pi)\, h(x_{\ell_p})$ have positive coefficients; each is therefore bounded below by its coefficient times $\min_{x \in [x(b), x(a)]} h(x)$, giving a closed-form interval lower bound $\mathrm{LB}[a, b]$. With the pole term $-\frac{1}{4} e^{\varepsilon^2/4} \cosh(\log p/2) < 0$ and the archimedean side controlled by Theorem 4.4 (weighted total ≤ 2693), the interval criterion $S_3(53) + \mathrm{LB}[a, b] > 8\sqrt{\pi} b^3 \cdot 2693$ implies $B^{\infty}_{\mathrm{GW}}(53, \varepsilon) < 0$ on $[a, b] \cap (0, \frac{1}{4}]$. Here the archimedean constant 2693 is the weighted total $\lceil \frac{I_4(1/4)}{2\pi} \sum_{p\le 53} (\log p)^{-2} \rceil = \lceil \frac{3561.1}{2\pi} \cdot 4.7514 \rceil$, with $I_4(\frac{1}{4}) \le 3561.1$ established through the explicit integration-by-parts bounds of Lemma 4.3; this bound is *recertified in Arb ball arithmetic* by the interval certificate (`certify_I4`), which outputs $I_4(\frac{1}{4}) \le 2408.17$ and hence $G_w \le 2693$, so 2693 is a certified output, not a numerical input. Evaluating the exact finite sums at 256-bit precision gives the two stated single-interval certificates with margins 0.83 and 24.29. □

Remark (Certificate data)**.** *The certified inequalities of Theorem 6.2 were produced by the interval certificate in the repository [4], run in Arb ball arithmetic (FLINT 3, Python bindings) with directed outward (midpoint–radius) rounding on all operations at 256-bit working precision (≈77 digits). All interval endpoints are decimal rationals. The two reference intervals cover* $[10^{-9}, 0.047]$ *(margin* 0.83*) and* $[0.047, 0.05]$ *(margin* 24.29*), i.e. the interval certificate covers* $[10^{-9}, 0.05]$*; the remaining* $(0, 10^{-9})$ *is covered by the closed-form Theorem 6.1 (valid for* $\varepsilon \le 0.004$*), so the* union *of the two arguments gives* $(0, 0.05]$*. The extension to* $\varepsilon \le 0.0819$ *uses* 378 *chained intervals of the same form. In each interval the finite sum runs over all prime powers with* $|\log n \mp \log p| < 3$*, the discarded tail being rigorously below* 10^{-380}*. The per-interval lower bound, right side, and margin are tabulated in the script output, archived under the release tag of [4]. The certified run used* `PREC_BITS` $= 256$*,* $\kappa = 53$*,* `GW_ARCH_BOUND` $= 2693$*,* `TAIL_CUTOFF` $= 3$*, covering intervals* $[10^{-9}, 0.047]$ *and* $[0.047, 0.05]$*, parameter-hash* `2b405227`*, source-hash* `d080b67e` *(the self-hash of the certificate script, regenerated at the release tag). The certified margins are* 0.8345 *and* 24.2883*.*

Corollary 6.3 (Full-object curvature positivity; proved)**.** *The full Guinand–Weil second-moment curvature is positive,* $-2B^{\infty}_{\mathrm{GW}}(53, \varepsilon) > 0$ *for all* $0 < \varepsilon \le 0.05$*. The operator curvature* $O''(\frac{1}{2}) = -2B^{\infty}_{\mathrm{line}}$ *is obtained from this once the off-line correction* $E_p^{\mathrm{off},(2)}$ *is bounded; see Corollary 8.3 after Theorem 8.2.*

Remark (Numerical illustration at the reference parameter)**.** *The proved sign* $B^{\infty}_{\mathrm{GW}} < 0$ *is made concrete through the on-line proxy at* $(\kappa, \varepsilon, N) = (53, 0.05, 100)$*, purely as an illustration; the proof above does not rely on this evaluation. The prime side is dominated by the diagonal eigenterms* $\log p/(\sqrt{p}\, 8\sqrt{\pi}\, \varepsilon^3)$*, summing to* $S_3(53) = 87.4247$ *before cross-prime corrections, the tightest of which sits at the near-zone gap* $\delta_{\min}(53) = 0.031749$ *between* $\log 32$ *and* $\log 31$ *(the prime power* 2^5 *against the prime* 31*). Each pole term* $-\frac{1}{4} e^{\varepsilon^2/4} \cosh(\frac{1}{2}\log p)$ *is negative, and the archimedean budget is bounded by* 2693 *(with true weighted value* ≈ 5.0*). The net second-moment curvature sum evaluates to* $B(53, 0.05, 100) = -19\,342.5$*, hence* $O''(\frac{1}{2}) = +38\,685.1$*, in agreement with the certified sign. The illustration uses the first* $N = 100$ *ordinates; the theorems themselves are parameter-symbolic and do not depend on a finite zero list.*

Remark (Certified continuation)**.** *Chaining* 378 *analytic interval certificates of the same type extends the certified range to* $\varepsilon \le 0.0819$*, where the exact margin crosses zero against the present constants. Above* $\varepsilon \approx 0.07$ *the binding term is the archimedean budget (true weighted value* ≈ 5.0 *versus the bound* 2693*); a sharper archimedean constant would extend certification further. The certified sign* $B^{\infty}_{\mathrm{GW}}(53, \varepsilon) < 0$ *holds throughout, and the on-line proxy value is negative numerically for all tested* $\varepsilon \in [0.005, 0.60]$*, with* $B(53, 0.05, 100) = -19\,342.5$*.*

7 Uniformity in the Cutoff

For general κ let $\delta_{\min}(\kappa)$ denote the minimal distance $|\log p - m\log q|$ over primes $p \le \kappa$ and prime powers $q^m \neq p$ (with no restriction $q^m \le \kappa$).

Lemma 7.1 (Elementary gap bound; proved)**.** *For all $\kappa \ge 2$, $\delta_{\min}(\kappa) \ge \dfrac{1}{e\kappa}$.*

Proof. Only pairs with $|\log p - m\log q| < 1$ matter, i.e. $q^m \in (p/e, ep) \subset (0, e\kappa)$. Since p is prime and $q^m \neq p$, the integers p and q^m are distinct, so $|p - q^m| \ge 1$; the mean value theorem for log gives $|\log p - m\log q| = |p - q^m|/\xi \ge 1/(e\kappa)$ with $\xi < e\kappa$. □

Lemma 7.2 (Global constant assembly; proved)**.** *With the Chebyshev-type inputs $\vartheta(x) > 0.84x$ ($x \ge 101$), $\psi(x) \le 1.04x$ and $\pi(x) < 1.25506\,x/\log x$ of [21], for $\kappa \ge 202$:*

(a) $S_3(\kappa) \ge 0.32\sqrt{\kappa}\,(\log\frac{\kappa}{2})^2$;

(b) $W_{\mathrm{near}}(\kappa) := \sum_{p\le\kappa}(\log p)^2 \sum_{(q,m)\in\mathcal{N}_p} \frac{\log q}{q^{m/2}} \le 5.85\,\kappa^{3/2}\log\kappa$ *(the direct near-zone sum,* not $\sum_p n_p \max$*);*

(c) $G_w(\kappa) \le 1481\,\kappa/\log\kappa$;

(d) $|B_{\mathrm{pol}}(\kappa,\varepsilon)| \le 0.32\,\kappa^{3/2}\log\kappa$ *for* $\varepsilon \le \frac{1}{4}$.

Moreover each ratio of the four error classes to $S_3(\kappa)/4$ is strictly decreasing on $[202,\infty)$.

Proof. (a) Restricting to $\kappa/2 < p \le \kappa$ and using $\sqrt{p} > \sqrt{\kappa/2}$, $\log p > \log(\kappa/2)$,

$$S_3(\kappa) \ge (\log\tfrac{\kappa}{2})^2\,\kappa^{-1/2}\big(\vartheta(\kappa) - \vartheta(\kappa/2)\big) \ge (\log\tfrac{\kappa}{2})^2\kappa^{-1/2}\big(0.84\kappa - 1.04\cdot\tfrac{\kappa}{2}\big) = 0.32\sqrt{\kappa}(\log\tfrac{\kappa}{2})^2,$$

since $0.84 - 0.52 = 0.32$ and $\kappa/2 \ge 101$. *(b)* For fixed p the near zone is the window $|L - m\log q| < 1$ of Theorem 5.1, i.e. every near prime power lies in $(p/e, ep)$, so the *direct* sum $\sum_{(q,m)\in\mathcal{N}_p}\frac{\log q}{q^{m/2}} = \sum\Lambda(n)/\sqrt{n} \le e^{1/2}p^{-1/2}\psi(pe) \le 1.04\,e^{3/2}\sqrt{p}$ (bounding the sum itself, with no n_p max inflation); then $\sum_{p\le\kappa}(\log p)^2\sqrt{p} \le \sqrt{\kappa}\,(\log\kappa)^2\,\pi(\kappa) \le \sqrt{\kappa}\,(\log\kappa)^2\cdot 1.25506\,\kappa/\log\kappa = 1.25506\,\kappa^{3/2}\log\kappa$, and $1.04\,e^{3/2}\cdot 1.25506 = 5.849 \le 5.85$, so $W_{\mathrm{near}} \le 5.85\kappa^{3/2}\log\kappa$. *(c)* $\sum_{p\le\kappa}\ell_p^{-2} \le \pi(\kappa)/(\log 2)^2$ with $\Gamma_p^{(2)} = O(1)$ from Theorem 4.4: the per-prime archimedean weight is $\le I_4(\frac14)/2\pi \le 566.77$, and $566.77\cdot 1.25506/(\log 2)^2 \le 1481$. *(d)* $\cosh(\ell_p/2) \le \sqrt{p}$ and $\frac14 e^{1/64}\cdot 1.25506 = 0.3187 \le 0.32$. *Monotonicity.* Writing $u = \log\kappa$, $v = \log(\kappa/2) = u - \log 2$, the four ratios are $R_{\mathrm{near}} \propto u/(\kappa v^2)$, $R_{\mathrm{arch}} \propto \kappa^{-5/2}u^{-5/2}v^{-2}$, $R_{\mathrm{pole}} \propto \kappa^{-2}u^{-1/2}v^{-2}$, and R_{far} carries the factor $e^{-e^2\kappa^2 u/2}$. Their logarithmic derivatives are

$$\tfrac{d}{d\kappa}\log R_{\mathrm{near}} = \tfrac{1}{\kappa u} - \tfrac{1}{\kappa} - \tfrac{2}{\kappa v} = -0.0062 < 0, \quad \tfrac{d}{d\kappa}\log R_{\mathrm{arch}} = -\tfrac{2}{\kappa} - \tfrac{5}{2\kappa u} - \tfrac{2}{\kappa v} = -0.0144 < 0,$$

$$\tfrac{d}{d\kappa}\log R_{\mathrm{pole}} = -\tfrac{2}{\kappa} - \tfrac{1}{2\kappa u} - \tfrac{2}{\kappa v} = -0.0125 < 0 \quad (\text{all at } \kappa = 202),$$

each manifestly negative for all $\kappa \ge 202$ since every summand after the first is negative and the first, $1/(\kappa\log\kappa)$, is dominated by $1/\kappa$; R_{far} decreases trivially. Hence all four ratios decrease on $[202,\infty)$. □

Theorem 7.3 (Uniform negativity threshold; proved)**.** *Let* $\varepsilon_0(\kappa) := \dfrac{1}{4e\kappa\sqrt{\log\kappa}}$. *Then for every* $\kappa \geq 202$ *and every* $0 < \varepsilon \leq \varepsilon_0(\kappa)$,

$$B_{\mathrm{GW}}^{\infty}(\kappa, \varepsilon) \;<\; 0.$$

Proof. Fix $\kappa \geq 202$ and $\varepsilon \leq \varepsilon_0 := \varepsilon_0(\kappa)$. We show each of the four error classes in $R(\varepsilon)$ is below $S_3(\kappa)/4$ at ε_0; since every bound below is increasing in ε, this covers the whole interval. By Lemma 7.2, $S_3(\kappa) \geq 0.32\sqrt{\kappa}(\log\frac{\kappa}{2})^2$ and the assembled weighted constants $W_{\mathrm{near}}(\kappa) \leq 5.85\,\kappa^{3/2}\log\kappa$, $G_w(\kappa) \leq 1481\,\kappa/\log\kappa$, $|B_{\mathrm{pol}}(\kappa,\varepsilon)| \leq 0.32\,\kappa^{3/2}\log\kappa$.

Near zone. By Lemma 7.1, every near pair has $x = \delta^2/4\varepsilon_0^2 \geq \delta_{\min}^2/4\varepsilon_0^2 \geq 4\log\kappa \geq \frac{1}{2}$, so $(1+2x)e^{-x} \leq 2.95\,\kappa^{-2}$, whence Near $\leq 17.3\,\kappa^{-1/2}\log\kappa$ (the value 6.46 at $\kappa = 202$).

Far zone. Far $\leq 20.5\,\kappa^{3/2}\log\kappa\; e^{-e^2\kappa^2\log\kappa/2}$ with $e^{-1/(32\varepsilon_0^2)} = e^{-e^2\kappa^2\log\kappa/2}$; at $\kappa = 202$ this is below 10^{-100} and stays $< 0.08\sqrt{\kappa}(\log\frac{\kappa}{2})^2$ for all $\kappa \geq 2$.

Archimedean. $8\sqrt{\pi}\varepsilon_0^3 G_w(\kappa) \leq 16.4/(\kappa^2(\log\kappa)^{5/2})$, which equals $6.2\cdot 10^{-6}$ at $\kappa = 202$ and is decreasing in κ.

Pole. $8\sqrt{\pi}\varepsilon_0^3|B_{\mathrm{pol}}| \leq 3.53\cdot 10^{-3}\,\kappa^{-3/2}(\log\kappa)^{-1/2}$, which equals $5.4\cdot 10^{-7}$ at $\kappa = 202$ and is decreasing in κ.

At the endpoint $\kappa = 202$ each of the four error classes is below $S_3(\kappa)/4 \geq 0.08\sqrt{\kappa}(\log\frac{\kappa}{2})^2 = 24.2$ (near 6.46, far $< 10^{-100}$, archimedean $6.2\cdot 10^{-6}$, pole $5.4\cdot 10^{-7}$), so their sum is below $S_3(202)$. The ratio of each class to $S_3(\kappa)/4$ is decreasing for $\kappa \geq 202$, so the comparison persists:

$$\frac{\text{Near}}{S_3/4} \leq \frac{216.25\,\log\kappa}{\kappa(\log\frac{\kappa}{2})^2} \quad (= 0.2668 \text{ at } \kappa = 202), \qquad \frac{\text{Far}}{S_3/4} \leq \frac{257\,\kappa\log\kappa}{(\log\frac{\kappa}{2})^2}e^{-e^2\kappa^2\log\kappa/2},$$

$$\frac{8\sqrt{\pi}\varepsilon_0^3 G_w}{S_3/4} \leq \frac{205}{\kappa^{5/2}(\log\kappa)^{5/2}(\log\frac{\kappa}{2})^2}, \qquad \frac{8\sqrt{\pi}\varepsilon_0^3|B_{\mathrm{pol}}|}{S_3/4} \leq \frac{0.045}{\kappa^2(\log\kappa)^{1/2}(\log\frac{\kappa}{2})^2},$$

each ratio decreasing on $[202,\infty)$ by the logarithmic-derivative computation of Lemma 7.2.

The monotonicity in ε and the condition $x \geq \frac{1}{2}$ hold on all of $(0, \varepsilon_0]$ because $\varepsilon_0 \leq \delta_{\min}/\sqrt{2}$; hence $-8\sqrt{\pi}\varepsilon^3 B_{\mathrm{GW}}^{\infty} \geq S_3 - \text{Near} - \text{Far} - 8\sqrt{\pi}\varepsilon^3(G_w + |B_{\mathrm{pol}}|) > 0$ throughout. □

Corollary 7.4 (Uniform full-object curvature positivity; proved)**.** *For every* $\kappa \geq 202$ *and* $0 < \varepsilon \leq \varepsilon_0(\kappa)$, *the full Guinand–Weil curvature is positive,* $-2B_{\mathrm{GW}}^{\infty}(\kappa,\varepsilon) > 0$. *The operator curvature* $O''(\frac{1}{2}) = -2B_{\mathrm{line}}^{\infty}$ *follows in the certified range* $\varepsilon_{\mathrm{off}} \leq \varepsilon \leq \varepsilon_0(\kappa)$ *by the off-line transfer of Theorem 8.2 (Corollary 8.3).*

Remark (Calibration)**.** *The threshold* $\varepsilon_0(\kappa)$ *is conservative by design. Numerically* $\kappa\,\delta_{\min}(\kappa) \in [1.6, 2.5]$ *over* $\kappa \in [53, 3203]$, *so Lemma 7.1 is sharp up to the constant; at* $\kappa \in \{101, 211\}$ *the full exact criterion at* $\varepsilon = \varepsilon_0(\kappa)$ *holds with margins* 180.6 *and* 396.1 *(essentially the full main term).*

8 Off-Line Robustness

Sections 3–7 are proved on the prime/pole/archimedean side and assume nothing about zero locations. We now isolate and bound the contribution of any hypothetical off-critical zeros to the zero-side sum $Z^{(2)}_{p,\mathrm{GW}} = Z^{(2)}_{p,\mathrm{line}} + E_p^{\mathrm{off},(2)}$, where $E_p^{\mathrm{off},(2)}(\varepsilon) = \frac{1}{2}\sum_{\rho:\beta\neq 1/2} h^{(2)}_{p,\varepsilon}\big((\rho-\frac{1}{2})/i\big)$.

Lemma 8.1 (Quadruple form and β-uniform magnitude bound; proved)**.** *Off-critical zeros occur in quadruples $Q = \{\rho, \bar\rho, 1-\rho, 1-\bar\rho\}$ with $\rho = \beta + i\gamma$, $\delta := \beta - \frac{1}{2} \neq 0$. The quadruple contribution is real,* $\mathrm{contrib}_Q(\varepsilon) = 2\,\Re\, h^{(2)}_{p,\varepsilon}(z_1)$ *with $z_1 = \gamma - i\delta$, and satisfies, independently of β,*

$$\big|\mathrm{contrib}_Q(\varepsilon)\big| \le 2\big(\gamma^2 + \tfrac{1}{4}\big)e^{\varepsilon^2/4}\sqrt{p}\, e^{-\varepsilon^2\gamma^2}.$$

Proof. Evenness gives $h^{(2)}(-\gamma \mp i\delta) = h^{(2)}(\gamma \pm i\delta)$, and real Taylor coefficients give $h^{(2)}(\gamma + i\delta) = \overline{h^{(2)}(z_1)}$, so the quadruple sum collapses to $2\Re h^{(2)}(z_1)$ after the zero-side factor $\frac{1}{2}$. Then $|z_1|^2 = \gamma^2 + \delta^2 \le \gamma^2 + \frac{1}{4}$, $|e^{-\varepsilon^2 z_1^2}| = e^{-\varepsilon^2(\gamma^2-\delta^2)} \le e^{-\varepsilon^2\gamma^2}e^{\varepsilon^2/4}$, and $|\cos(z_1 \log p)| \le e^{|\delta|\log p} = p^{|\delta|} < \sqrt{p}$, using $|\delta| < \frac{1}{2}$. ☐

Theorem 8.2 (Off-line robustness, magnitude form; proved)**.** *Let κ be the prime cutoff, $H_0 = 3 \cdot 10^{12}$, and $f_\varepsilon(t) := (t^2 + \frac{1}{4})e^{-\varepsilon^2 t^2}$, which is decreasing on $[H_0, \infty)$ once $\varepsilon H_0 \ge 2$. Then*

$$\sum_{p\le\kappa}(\log p)^2\big|E_p^{\mathrm{off},(2)}(\varepsilon)\big| \;\le\; 2\,C_\kappa\, e^{\varepsilon^2/4}\, S^{\mathrm{disc}}_{\mathrm{tail}}(\varepsilon, H_0), \qquad C_\kappa = \sum_{p\le\kappa}(\log p)^2\sqrt{p},$$

where the discrete tail

$$S^{\mathrm{disc}}_{\mathrm{tail}}(\varepsilon, H_0) := \sum_{n\ge 0}\big(N(H_0+n+1) - N(H_0+n)\big)\, f_\varepsilon(H_0+n)$$

uses, for each bin count, the explicit Hasanalizade–Shen–Wong inequality $|N(T) - F(T)| \le E(T)$ of [22], with

$$F(T) = \tfrac{T}{2\pi}\log\tfrac{T}{2\pi e}, \qquad E(T) = 0.1038\log T + 0.2573\log\log T + 9.3675,$$

so that each bin count satisfies the explicit estimate

$$N(H_0+n+1) - N(H_0+n) \;\le\; F(H_0+n+1) - F(H_0+n) + E(H_0+n+1) + E(H_0+n).$$

We use the Hasanalizade–Shen–Wong constants because they are sufficient for the present threshold and match the archived certificate. Newer explicit zero-count estimates of Bellotti–Wong [23] improve these constants and could slightly sharpen the numerical value of $\varepsilon_{\mathrm{off}}$, but are not needed for the proof. Consequently, for each certified negativity margin M there is an explicit threshold $\varepsilon_{\mathrm{off}}(\kappa, M, H_0)$ with $\sum_{p\le\kappa}(\log p)^2|E_p^{\mathrm{off},(2)}(\varepsilon)| < M$ for all $\varepsilon \ge \varepsilon_{\mathrm{off}}$. Here M is the scaled *interval-criterion margin of Theorem 6.2 (e.g.* 0.83 *or* 21.86*); since $8\sqrt{\pi}\varepsilon^3 < 1$ on $(0, 0.05]$, it lies below the unscaled negativity margin $-B^\infty_{\mathrm{GW}} = (8\sqrt{\pi}\varepsilon^3)^{-1}\cdot$(scaled gap), so $\sum_{p\le\kappa}(\log p)^2|E_p^{\mathrm{off},(2)}| < M$ is a conservative sufficient condition for $B^\infty_{\mathrm{line}} < 0$. For $\kappa = 53$, $C_{53} = 782.89$. The proved closed-form majorant above* exists *as an analytic threshold; its concrete decimal value is ob-*

tained by numerical evaluation (mpmath, not Arb-interval-certified). Against the smallest reference margin $M = 0.83$ *it gives* $\varepsilon_{\text{off}}(53, 0.83, H_0) \le 3.20 \cdot 10^{-12}$*, and against the uniform margin* $M = S_3/4 = 21.86$*,* $\varepsilon_{\text{off}}(53, 21.86, H_0) \le 3.14 \cdot 10^{-12}$ *(these values are* $\kappa = 53$*-specific, not uniform in* κ*, and numerical); the independent numerical discrete evaluation gives* $3.15 \cdot 10^{-12}$ *and* $3.09 \cdot 10^{-12}$ *respectively, the closed form agreeing to within* 1% *(the gap is the slowly-varying bin-density bound). Hence, on a certified full-object negativity range with margin* M *and for* $\varepsilon \ge \varepsilon_{\text{off}}(\kappa, M, H_0)$ *— that is, for* $\varepsilon_{\text{off}}(\kappa, M, H_0) \le \varepsilon \le 0.05$*, in particular at* $\varepsilon = 0.05$ *(ten orders of magnitude above* ε_{off}*) — the certified sign of* B^{∞}_{GW} *transfers to the on-line proxy:* $B^{\infty}_{\text{line}}(53, \varepsilon) < 0$*.*

Proof. Step 1 (finite verified range). By Platt–Trudgian [24] there is no off-critical zero with $0 < \gamma \le H_0$ (the verified height is 3 000 175 332 800, above $3 \cdot 10^{12}$), so $\sum_p (\log p)^2 |E_p^{\text{off},(2)}|$ reduces to off-critical quadruples with $\gamma > H_0$.

Step 2 (off-line support and discrete bin majorant). By the object split, $E_p^{\text{off},(2)} = Z^{(2)}_{p,\text{GW}} - Z^{(2)}_{p,\text{line}}$ receives no contribution from on-line zeros (there the full and proxy summands coincide), so $E_p^{\text{off},(2)}$ is supported on the off-critical quadruples Q of Lemma 8.1, each with $|\text{contrib}_Q| \le f_\varepsilon(\gamma)\sqrt{p}\, e^{\varepsilon^2/4}$. Writing N^{off} for the off-critical counting function, and using that the off-critical zeros in any bin are a subset of all zeros there ($N^{\text{off}}(H_0+n+1) - N^{\text{off}}(H_0+n) \le N(H_0+n+1) - N(H_0+n)$, with $f_\varepsilon \ge 0$),

$$\sum_{\substack{\gamma > H_0 \\ \beta \ne \frac{1}{2}}} f_\varepsilon(\gamma) \le \sum_{n \ge 0} \big(N(H_0+n+1) - N(H_0+n)\big)\, f_\varepsilon(H_0+n) = S^{\text{disc}}_{\text{tail}}(\varepsilon, H_0),$$

f_ε being decreasing on $[H_0, \infty)$ for $\varepsilon H_0 \ge 2$. This majorizes the off-line difference by the total zero count; the over-count (on-line zeros contribute 0 to $E_p^{\text{off},(2)}$ but are counted in N) is harmless for a magnitude statement and avoids any zero-density input.

Step 3 (zero-count bound). Each bin count $N(H_0+n+1) - N(H_0+n)$ is bounded by differencing the explicit inequality of [22]; this replaces the Riemann–von Mangoldt density and is a rigorous upper bound for a discrete zero sum. Multiplying by $2e^{\varepsilon^2/4}\sqrt{p}\,(\log p)^2$ and summing over $p \le \kappa$ gives the stated bound.

Step 4 (log-density summation and threshold comparison). The map $\varepsilon \mapsto 2C_\kappa e^{\varepsilon^2/4} S^{\text{disc}}_{\text{tail}}$ is strictly decreasing, and the threshold ε_{off} admits a *closed-form* majorant rather than a bare infinite-series bisection. The bin count is *not* uniformly bounded: $D_n := N(H_0+n+1) - N(H_0+n) = O(\log(H_0+n))$ grows with n. By the explicit HSW inequality and the concavity bound $\log(H_0+n) \le \log H_0 + n/H_0$ one has the affine majorant $D_n \le d_0 + d_1 n$ with the *secant* (not tangent) initial value $d_0 = F(H_0+1) - F(H_0) + E(H_0+1) + E(H_0)$, $F(t) = \frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}$, and $d_1 = \frac{1}{2\pi H_0} + 2E'(H_0)$, where $E'(H_0) = \frac{0.1038}{H_0} + \frac{0.2573}{H_0 \log H_0}$ captures the logarithmic growth of the HSW error term over the bins, so $d_1 \approx \frac{1}{2\pi H_0} + \frac{0.21}{H_0}$ — still negligible against $H_0 = 3 \cdot 10^{12}$. Keeping the full $f_\varepsilon(t) = (t^2 + \frac{1}{4})e^{-\varepsilon^2 t^2}$ (the $+\frac{1}{4}$ must not be dropped from an upper bound) and using, for $n \ge 0$,

$$f_\varepsilon(H_0+n) \le \big((H_0+n)^2 + \tfrac{1}{4}\big)\, e^{-\varepsilon^2 H_0^2}\, q^n, \qquad q := e^{-2\varepsilon^2 H_0} \in (0,1),$$

(from $e^{-\varepsilon^2(H_0+n)^2} \le e^{-\varepsilon^2 H_0^2} e^{-2\varepsilon^2 H_0 n}$ and $e^{-\varepsilon^2 n^2} \le 1$), the tail collapses to the geometric sums $\sum q^n, \sum nq^n, \sum n^2 q^n, \sum n^3 q^n$ (all closed forms in q):

$$S_{\mathrm{tail}}^{\mathrm{disc}}(\varepsilon, H_0) \le e^{-\varepsilon^2 H_0^2}\Big[d_0\big(\Sigma_2 + \tfrac14\Sigma_0\big) + d_1\big(\Sigma_2' + \tfrac14\Sigma_1\big)\Big],$$

with $\Sigma_0 = \sum q^n$, $\Sigma_1 = \sum nq^n$, $\Sigma_2 = \sum (H_0+n)^2 q^n$, $\Sigma_2' = \sum n(H_0+n)^2 q^n$, each a closed expression in q evaluable in ball arithmetic. The log-growth term $d_1 n$ changes only the prefactor; solving $2C_\kappa e^{\varepsilon^2/4} S_{\mathrm{tail}}^{\mathrm{disc}} = M$ yields the threshold directly. At $\varepsilon = 0.05$ the leading bin already carries the factor $f_{0.05}(H_0) = O\big(H_0^2 e^{-0.0025\, H_0^2}\big)$, far beneath every margin; the certified sign therefore transfers to $B_{\mathrm{line}}^{\infty}$, and with $O''(\frac12) = -2B_{\mathrm{line}}^{\infty}$ the operator curvature is positive in that range. □

Corollary 8.3 (Operator curvature positivity; proved)**.** Reference case $\kappa = 53$. *For every ε with $\varepsilon_{\mathrm{off}}(53, M, H_0) \le \varepsilon \le 0.05$ (in particular $\varepsilon = 0.05$, with the concrete thresholds $\varepsilon_{\mathrm{off}}(53, 0.83, H_0) \le 3.20 \cdot 10^{-12}$ and $\varepsilon_{\mathrm{off}}(53, 21.86, H_0) \le 3.14 \cdot 10^{-12}$, which are numerical and not Arb-certified, and are not needed at sharp precision since $\varepsilon = 0.05$ lies far above them), the on-line proxy curvature sum is negative, $B_{\mathrm{line}}^{\infty}(53, \varepsilon) < 0$, and hence the operator curvature is strictly positive:*

$$O''(\tfrac12) = -2B_{\mathrm{line}}^{\infty}(53, \varepsilon) > 0.$$

Uniform case $\kappa \ge 202$. *For every $\varepsilon_{\mathrm{off}}(\kappa, M, H_0) \le \varepsilon \le \varepsilon_0(\kappa)$ (where nonempty) one has $B_{\mathrm{line}}^{\infty}(\kappa, \varepsilon) < 0$ and $O''(\frac12) = -2B_{\mathrm{line}}^{\infty}(\kappa, \varepsilon) > 0$, by Theorem 7.3 and the transfer of Theorem 8.2.*

Proof. Theorems 6.2 and 7.3 give $B_{\mathrm{GW}}^{\infty} < 0$ in the stated ranges. For $\varepsilon \ge \varepsilon_{\mathrm{off}}$, Theorem 8.2 bounds the off-line sum below the certified (scaled) margin M, which satisfies $M < -B_{\mathrm{GW}}^{\infty}$ on the certified range. Since

$$B_{\mathrm{line}}^{\infty} = B_{\mathrm{GW}}^{\infty} - \sum_{p \le \kappa} (\log p)^2 E_p^{\mathrm{off},(2)}, \qquad \Big|\sum_{p \le \kappa} (\log p)^2 E_p^{\mathrm{off},(2)}\Big| < M < -B_{\mathrm{GW}}^{\infty} = |B_{\mathrm{GW}}^{\infty}|,$$

the sign of $B_{\mathrm{line}}^{\infty}$ equals that of B_{GW}^{∞}, i.e. negative; $O''(\frac12) = -2B_{\mathrm{line}}^{\infty}$ then gives the curvature sign. □

Remark (What is and is not proved here)**.** *Theorem 8.2 is a* magnitude *statement: it shows that hypothetical off-critical zeros leave the negativity, and hence $O''(\frac12) > 0$, intact for every ε in a certified range with $\varepsilon \ge \varepsilon_{\mathrm{off}}$ (that is, $\varepsilon_{\mathrm{off}} \le \varepsilon \le 0.05$), in particular at the reference width $\varepsilon = 0.05$. It does not control the* sign *of $E_p^{\mathrm{off},(2)}$, and it does not prove RH. Since the negativity theorems are already established on the zero-location- free prime side, the distinct content is the explicit, β-uniform, finite-height certificate quantifying the gap between the on-line picture and the true zero configuration. Subsequent work has refined the complementary zero-free and zero-density regime above H_0 rather than the verified height itself ([26, 27]).*

9 Open Problems

Open Problem 9.1 (Cancellation Hypothesis). *Establish Assumption (A^{**}) of [3] analytically: a logarithmic saving in $|S_k(\kappa)|/P(\kappa)$ via Vinogradov–Korobov exponential-sum techniques (see [26, 27] for the state of the art on zero-free regions). Assumption (A^{**}) is a log-saving hypothesis, strictly weaker than the Riemann Hypothesis; no equivalence is claimed.*

Open Problem 9.2 (Weighted Bias Bridge). *Establish the bridge implication $B^{\infty}_{\mathrm{int}} < 0 \Rightarrow B^{\infty}_{\mathrm{GW}} < 0$ directly. The present paper does not use it: it proves $B^{\infty}_{\mathrm{GW}} < 0$ independently and so bypasses the bridge. Whether the integrated bias $B^{\infty}_{\mathrm{int}}$ controls the full Guinand–Weil object through such an implication remains open.*

Open Problem 9.3 (Exact stationarity). *Establish $O'(\frac{1}{2}) = 0$, or $|O'(\frac{1}{2})|/O''(\frac{1}{2}) \to 0$. The selection criterion of [3] quantifies the proximity to stationarity but not its exactness; establishing exactness remains open.*

Open Problem 9.4 (Weil-transfer operator and positivity). *For the prime-local antisymmetric Gaussian component $\psi_{p,\varepsilon}(x) = \phi_\varepsilon(x - \log p) - \phi_\varepsilon(x + \log p)$, with ϕ_ε the Gaussian $(2\pi\varepsilon^2)^{-1/2}\exp(-x^2/2\varepsilon^2)$, define the Weil-transfer matrix*

$$(\mathsf{W}^{\mathrm{Weil}}_{\kappa,\varepsilon})_{pq} := W(\psi_{p,\varepsilon} * \check{\psi}_{q,\varepsilon}),$$

where W is the Weil quadratic functional of [7, 14]. Establish whether a bridge identity relates $\langle \mathsf{W}^{\mathrm{Weil}}_{\kappa,\varepsilon} c^\eta, c^\eta\rangle$, with the coefficient vector $c^\eta = (\sqrt{V_p(\frac{1}{2})})_{p\le\kappa}$, to $O''(\frac{1}{2})$ up to off-diagonal and Γ-factor corrections, and whether the resulting form is positive on the growing finite-cutoff subspace. The vector c^η is not asserted to reproduce the $(\log p)^2$ curvature weights directly; identifying the exact renormalisation and correction terms between the c^η-weighted Weil form and the curvature weight is part of the problem. Positivity of the Weil form on this subspace would be a finite-cutoff analogue of the Weil positivity criterion of [7], which is known to be equivalent to the Riemann Hypothesis [7, 14].

Open Problem 9.5 (Scaling of the negativity window). *Determine the asymptotics of the negativity threshold; in particular, whether $\inf_\kappa \varepsilon_{\mathrm{int}}(\kappa) > 0$ for the integrated window of [3], and how the uniform threshold $\varepsilon_0(\kappa)$ of Theorem 7.3 compares with the observed scale $\delta_{\min}(\kappa) \asymp 2/\kappa$. Numerical evidence is oscillatory.*

Remark (Scope: the curvature is not a direct RH criterion). *The negativity $B^{\infty}_{\mathrm{GW}} < 0$ established here is a statement about the second-moment bias, and the operator curvature $O''(\frac{1}{2}) = -2B^{\infty}_{\mathrm{line}} > 0$ follows by the off-line transfer; neither is a criterion for the Riemann Hypothesis. The canonical off-critical continuation of the weighted curvature test function*

$$h^{(2,w)}_{p,\varepsilon}(u) := (\log p)^2 h^{(2)}_{p,\varepsilon}(u) = (\log p)^2 u^2 e^{-\varepsilon^2 u^2}\cos(u \log p),$$

written $G(\gamma) := h^{(2,w)}_{p,\varepsilon}(\gamma)$, is obtained as follows. A hypothetical off-critical zero $\rho = \frac{1}{2} + \delta + i\gamma$ enters the half-sum $B = \frac{1}{2}\sum_\rho$ together with its complex conjugate $\overline{\rho} = \frac{1}{2} + \delta - i\gamma$, contributing the arguments $u = (\rho - \frac{1}{2})/i = \gamma - i\delta$ and $u = (\overline{\rho} - \frac{1}{2})/i = -\gamma - i\delta$; since $h^{(2,w)}_{p,\varepsilon}$ is even in u,

$G(-\gamma - i\delta) = G(\gamma + i\delta)$, so this conjugate pair contributes $G(\gamma + i\delta) + G(\gamma - i\delta)$ to $\sum_\rho$, i.e. $\frac{1}{2}\big[G(\gamma + i\delta) + G(\gamma - i\delta)\big]$ in the half-sum, with on-line limit $G(\gamma)$ at $\delta = 0$. The defect relative to that limit is

$$\Delta B(\delta, \gamma) := \tfrac{1}{2}\big[G(\gamma + i\delta) + G(\gamma - i\delta) - 2G(\gamma)\big] = \tfrac{1}{2}\big[-\delta^2 G''(\gamma) + O(\delta^4)\big] = \delta^2 C_2(\gamma) + O(\delta^4),$$

the middle step being the even-order Taylor expansion $G(\gamma \pm i\delta) = G(\gamma) \mp \frac{\delta^2}{2} G''(\gamma) + O(\delta^4)$ along the imaginary direction. Hence

$$C_2(\gamma) = -\tfrac{1}{2} G''(\gamma),$$

the factor $-\frac{1}{2}$ originating in the half-sum $\frac{1}{2}\sum_\rho$; the functional-equation partners $1 - \rho, 1 - \overline{\rho}$ form an identical conjugate pair contributing the same defect, and the un-normalised convention $\Delta B := G(\gamma + i\delta) + G(\gamma - i\delta) - 2G(\gamma)$ (without the half-sum $\frac{1}{2}$) would instead read off $-G''(\gamma)$. The coefficient $C_2(\gamma)$ is sign-indefinite: for $p = 2$, $\varepsilon = 0.05$, $G''(\gamma)$ changes sign 13 times on $[1, 60]$ (e.g. $G''(5) \approx +6.41 > 0$ but $G''(9) \approx -14.85 < 0$), consistent with the larger-range numerical scans; equivalently, the Fourier transform of a second derivative is not of fixed sign. Hence the curvature bias does not furnish a Weil-positivity / RH criterion. This delimits the result and is stated for transparency.

Non-Circularity

We document explicitly that the results of this paper do not rely on the objects they are directed toward.

No RH as input. The Riemann Hypothesis is never assumed. All zero sums are defined through the zero side of the explicit formula and require no hypothesis on zero positions; the ordinate form quoted in § 2 coincides with this definition on the range of zeros entering any numerical computation. The results of §§ 3–7 are unconditional, and § 8 is a magnitude statement.

No GUE, Montgomery, or Hilbert–Pólya hypothesis. The distribution of the zeros is not modelled by random matrix theory. The operator $\widetilde{T}$ is not a Hilbert–Pólya operator ([9]: $r_2 \to 0.16$); its matrix entries are explicit deterministic arithmetic expressions. No pair-correlation conjecture is used.

Off-line robustness uses a verified height, not RH. The single external input of § 8 is the Platt–Trudgian computation [24], a rigorous unconditional interval-arithmetic verification that no off-critical zero exists below $H_0 = 3 \cdot 10^{12}$. This is a finite arithmetic fact, not an instance of RH.

$B^\infty_{\mathrm{GW}} < 0$ **is not assumed and not RH-equivalent.** The negativity is established by closed-form and certified computation (Theorems 6.1, 6.2, 7.3); it is a one-sided curvature/bias fact, and by the scope remark of § 9 it is not a criterion for RH. The non-vanishing $\zeta(1+it) \neq 0$ [29, 30] is an unconditional analytical input.

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Call for Feedback

This paper is part of an ongoing open research initiative toward the Riemann Hypothesis. All papers in the series are freely available on Zenodo under CC BY 4.0, and the accompanying verification code is hosted on GitHub.

Zenodo (https://zenodo.org/search?q=Ulrich+Tehrani) and GitHub (https://github.com/utehrani/analysislab-nt).

Topics of particular interest include: analytical approaches to Assumption (A**) via exponential-sum methods; the relation between the negativity window and zero-free regions for $\zeta(s)$; sharper archimedean constants; and any errors or gaps in the present arguments.

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